SHMtrace: revised figures

Codex design revision 2 · 22 October 2026

Seven main figures and the supplementary figure set, rebuilt as a separate copy.

Data figures use saved results; no caller defaults or statistical tests were changed.

Figure 3E–H moved to S25; Figure 4I moved to S26.

S8 spans two pages. S14–S16 comparison tables and S17 trees have dedicated pages.

Numbers are review identifiers. See the integration map before manuscript submission.

The original Word manuscript is unchanged. Updated captions and panel maps accompany this pack.

Editable SVG, Python sources, PNG, PDF and TIFF exports are in the package folder.

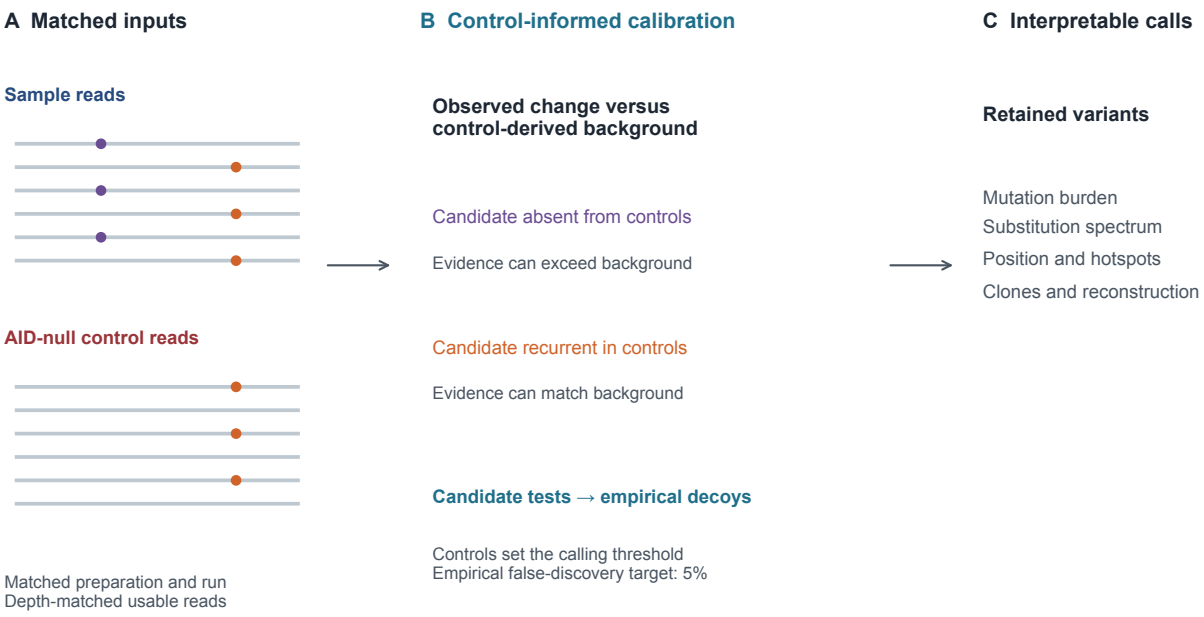
Review order

Main figures 1–7

Supplementary figures S1–S24, including companion pages

S25: downstream simulations; S26: read accounting

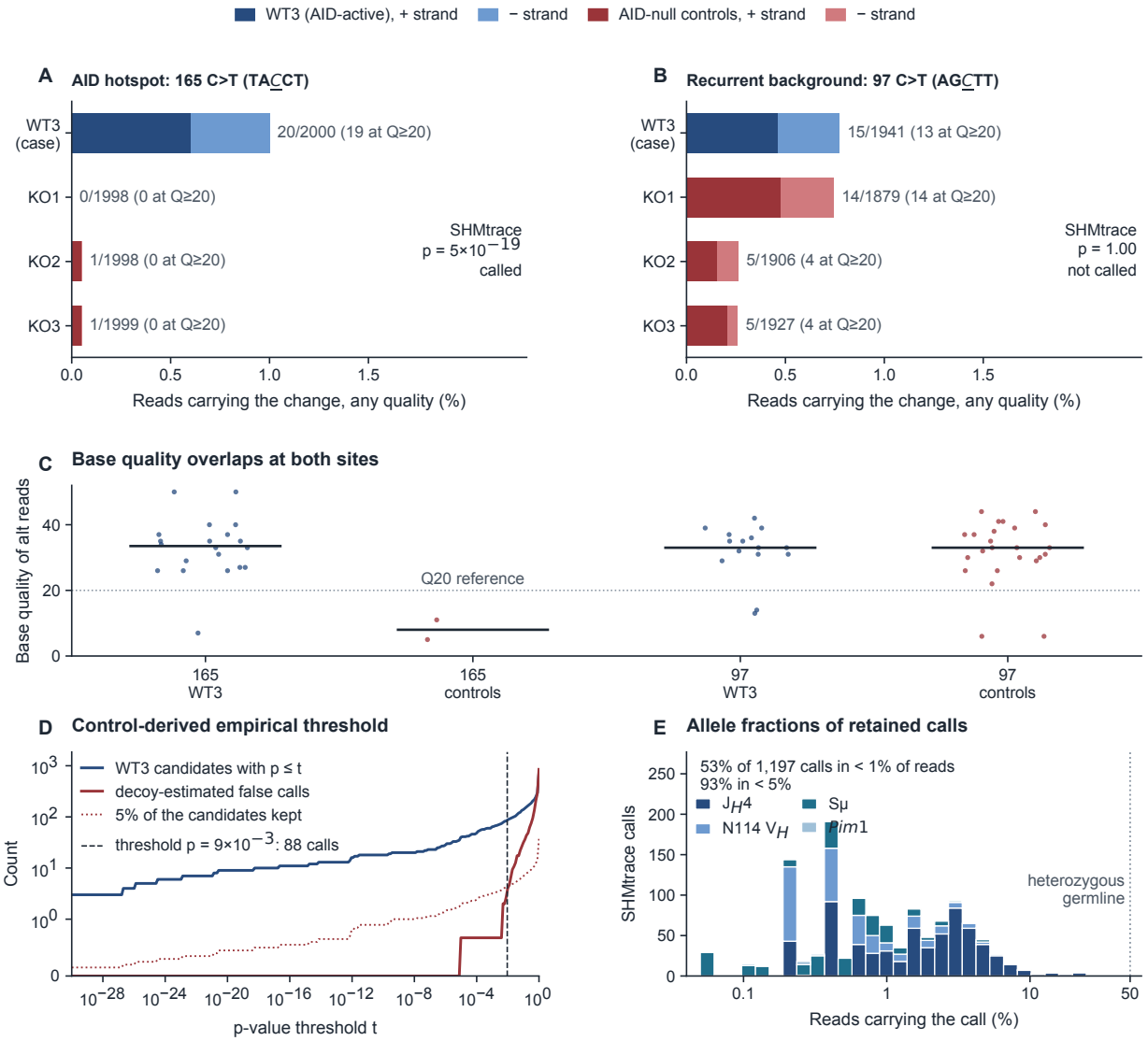
Figure 1. Control-informed mutation calling



Schematic pileups illustrate the decision logic; colours distinguish candidate changes and recurrent background.

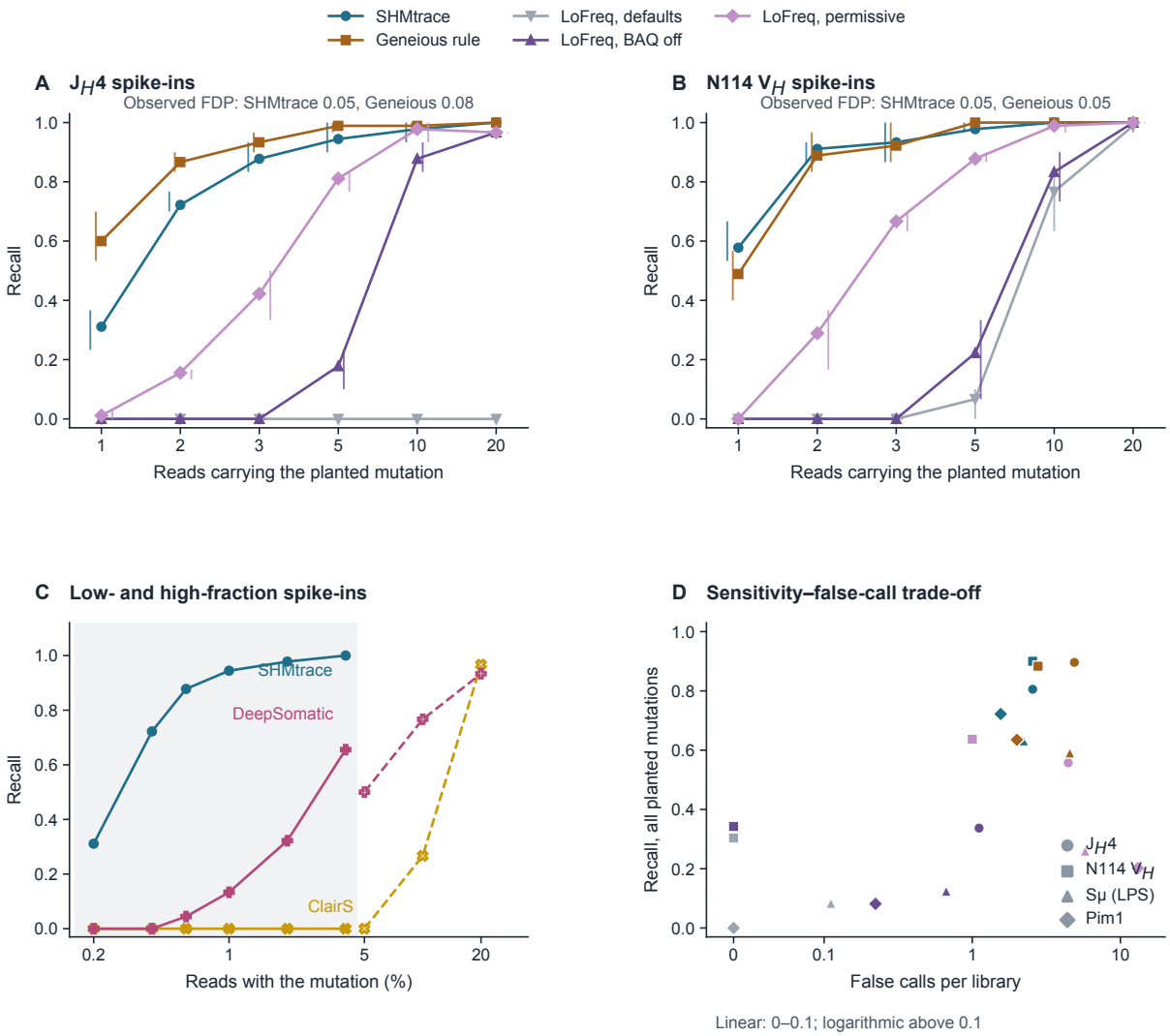
Open three-stage schematic; separate Sanger and PacBio measurement routes.

Figure 2. Matched controls distinguish candidates from background



Enlarged paired examples, full-width quality comparison, and uncluttered allele-fraction distribution.

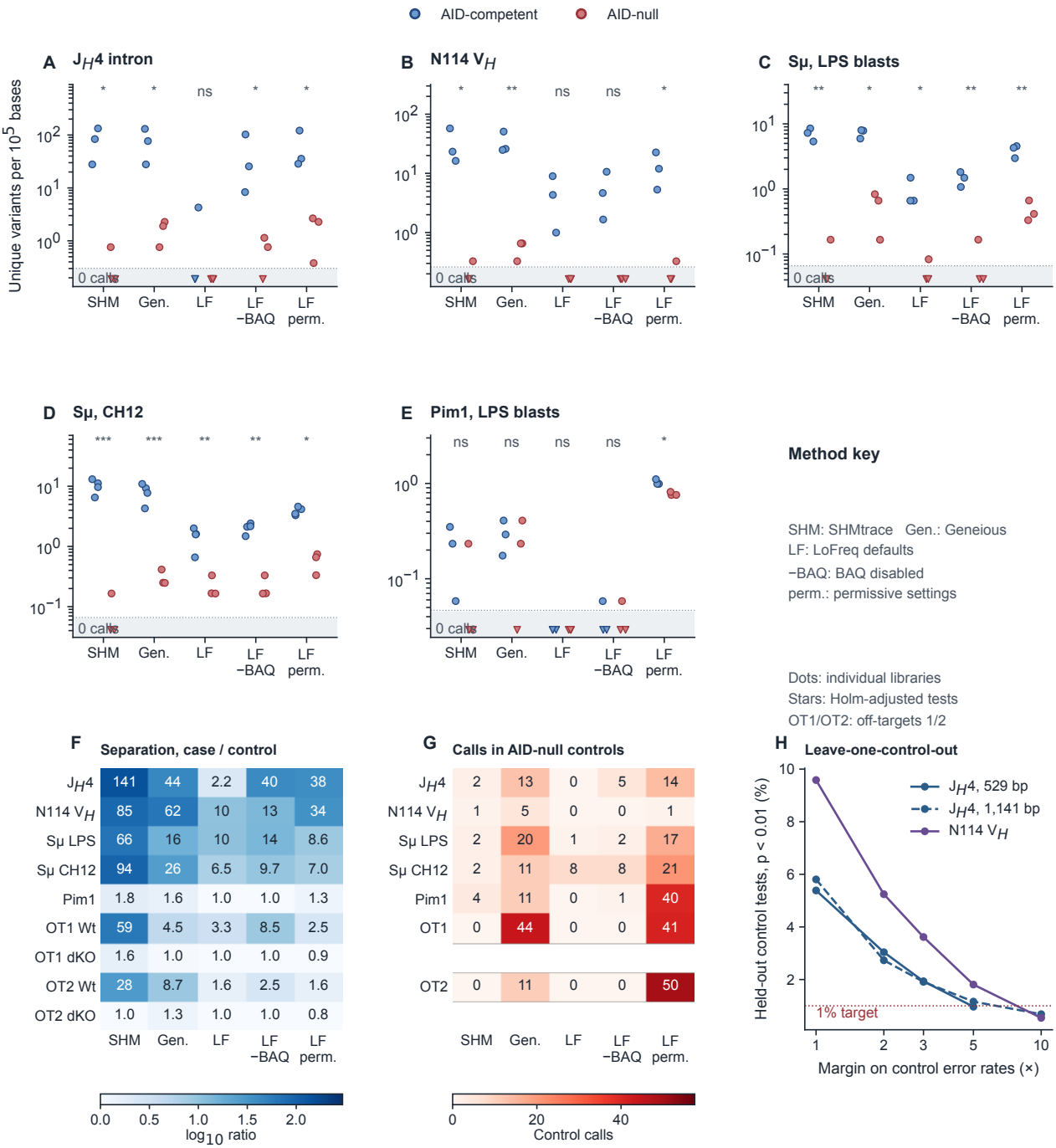
Figure 3. Sensitivity and false calls in spike-in benchmarks



Simulation-based quantification and group testing are shown separately in Figure S25.

Calling performance stays in the main figure; downstream simulations move to S25.

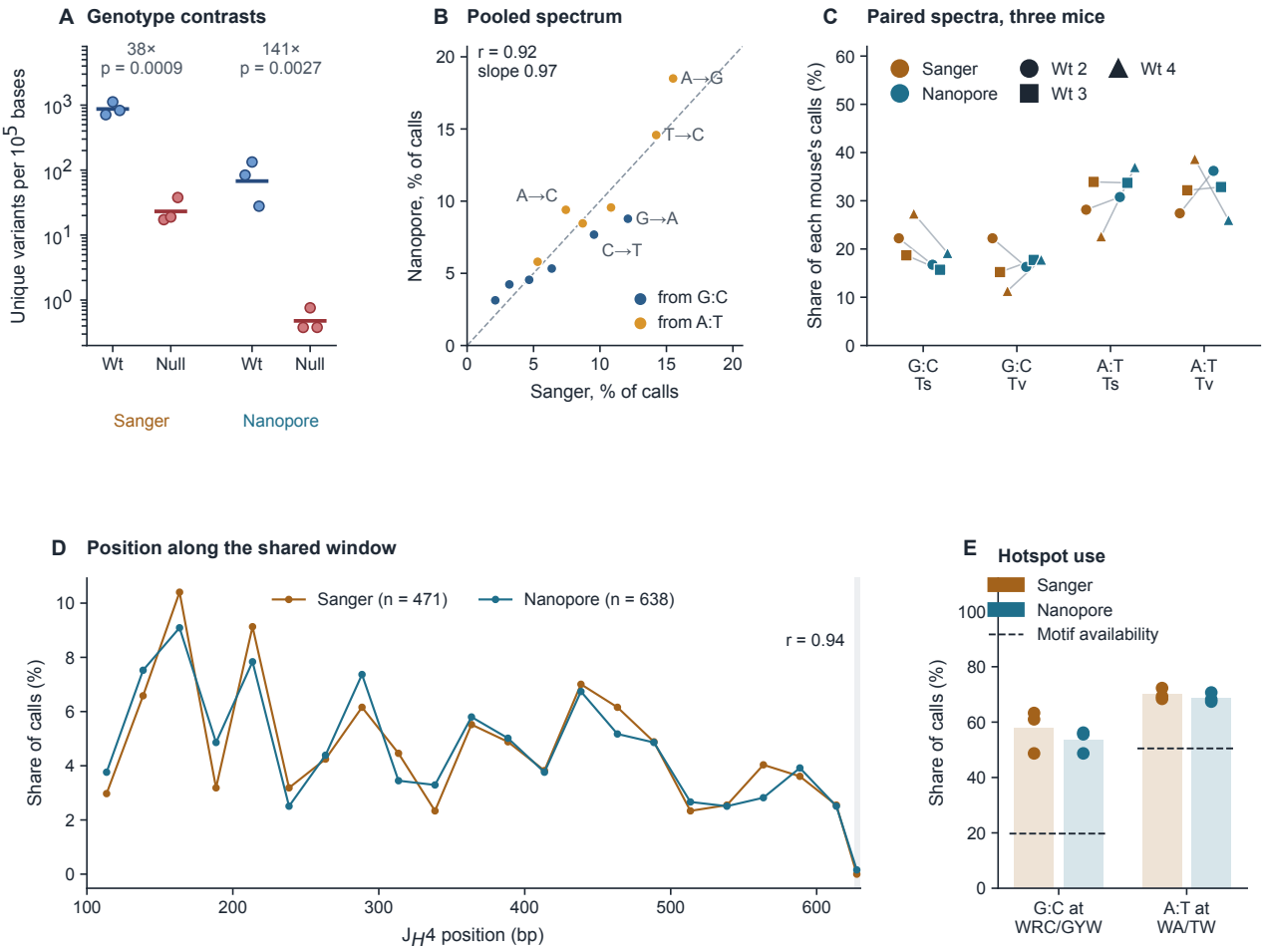
Figure 4. Matched controls in real libraries



F: geometric-mean case/control density ratio; zero-call libraries enter the ratio at one event/exposure.

Wider small multiples, zero-call strips and heatmap keys; read accounting moves to S26.

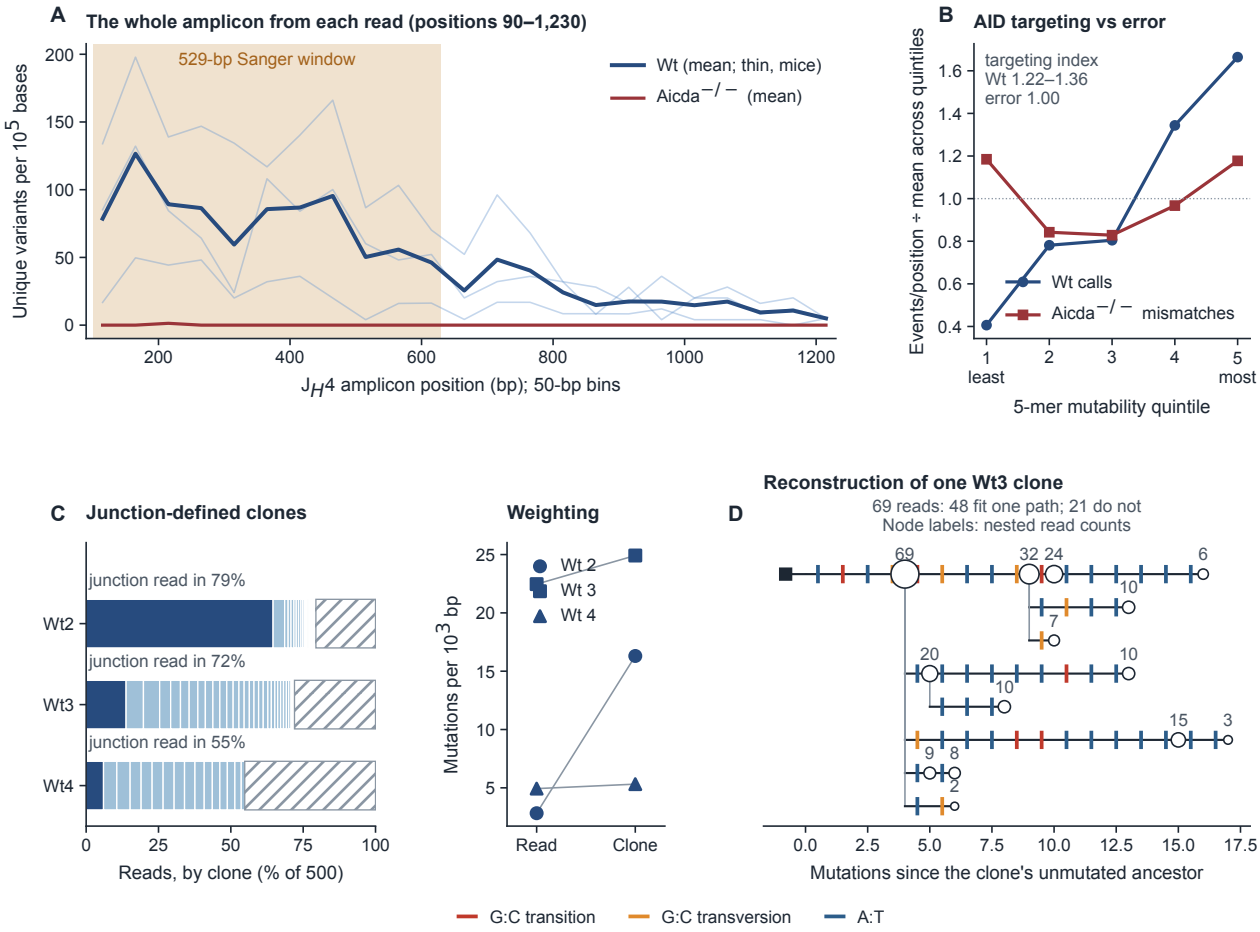
Figure 5. Matched Sanger and Nanopore mutation profiles



Pattern comparisons use within-platform shares; absolute densities count different units. Lines in C pair the same mouse.

Paired mouse observations replace dominant bars; motif availability is labelled explicitly.

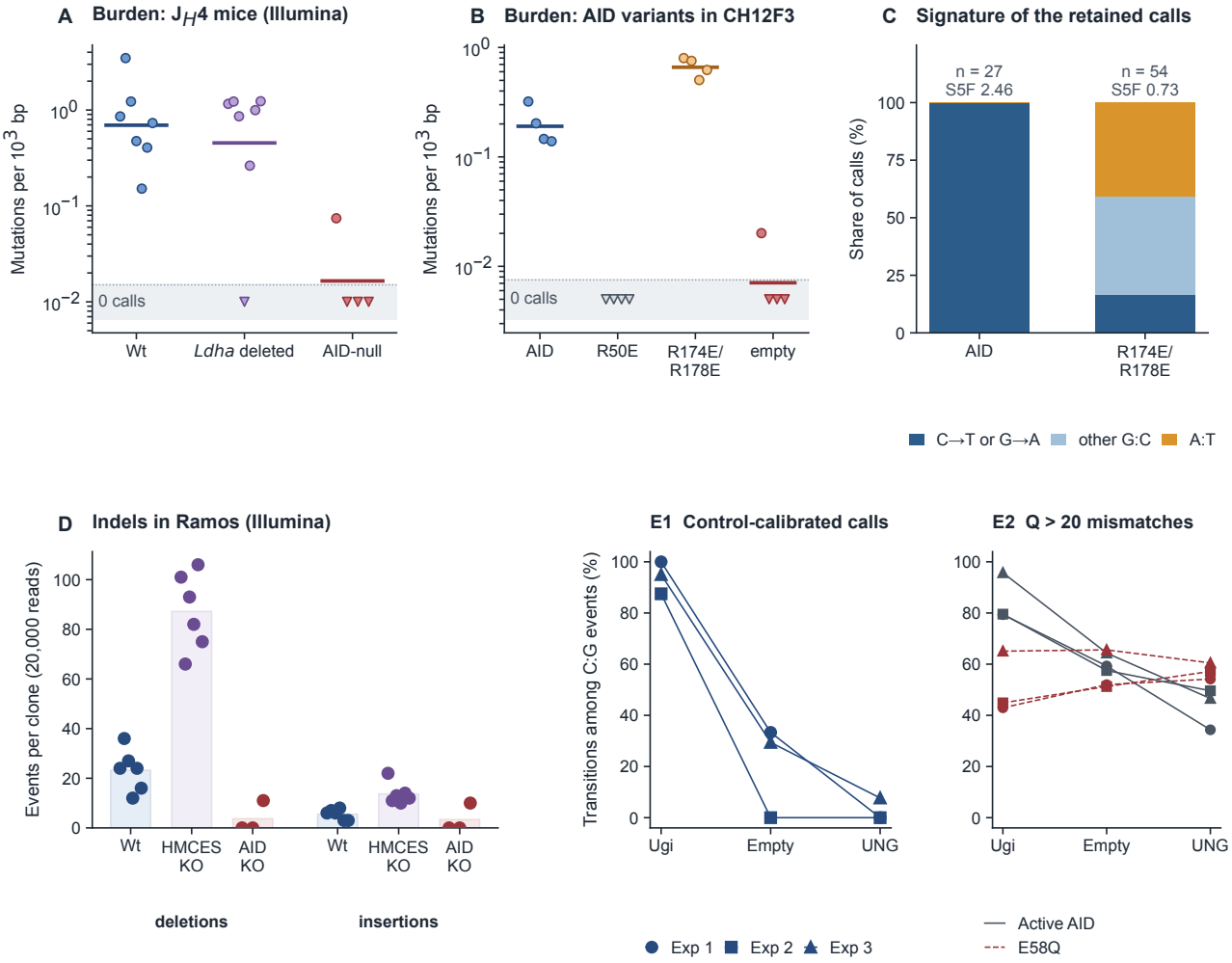
Figure 6. Full-length information and linked mutation patterns



C: hatched reads have no assigned junction. D: ticks mark mutation classes; circle area increases with read count. Branching is a reconstruction from observed sequences, not a directly measured genealogy.

Separated clone composition and weighting; clarified reconstruction and nested node counts.

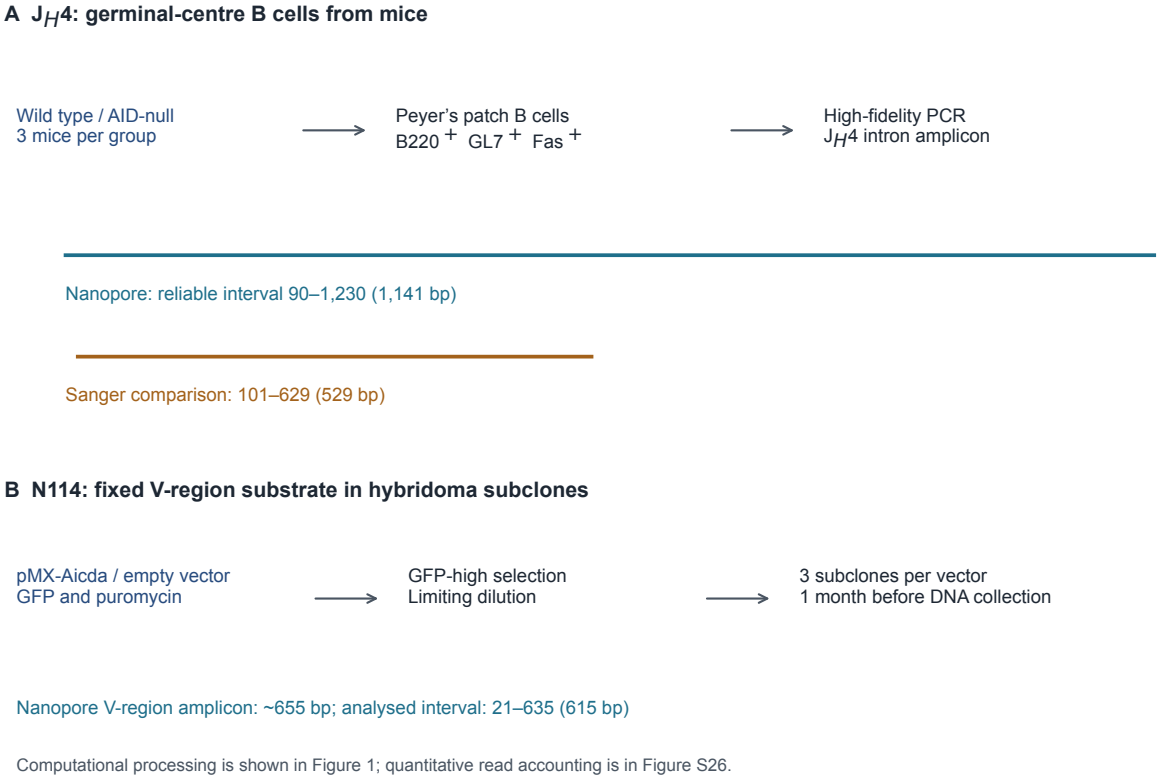
Figure 7. External datasets across biological contexts



E: Pérez-Durán non-B-cell dataset. Each shape is one experiment; lines connect its three repair conditions.

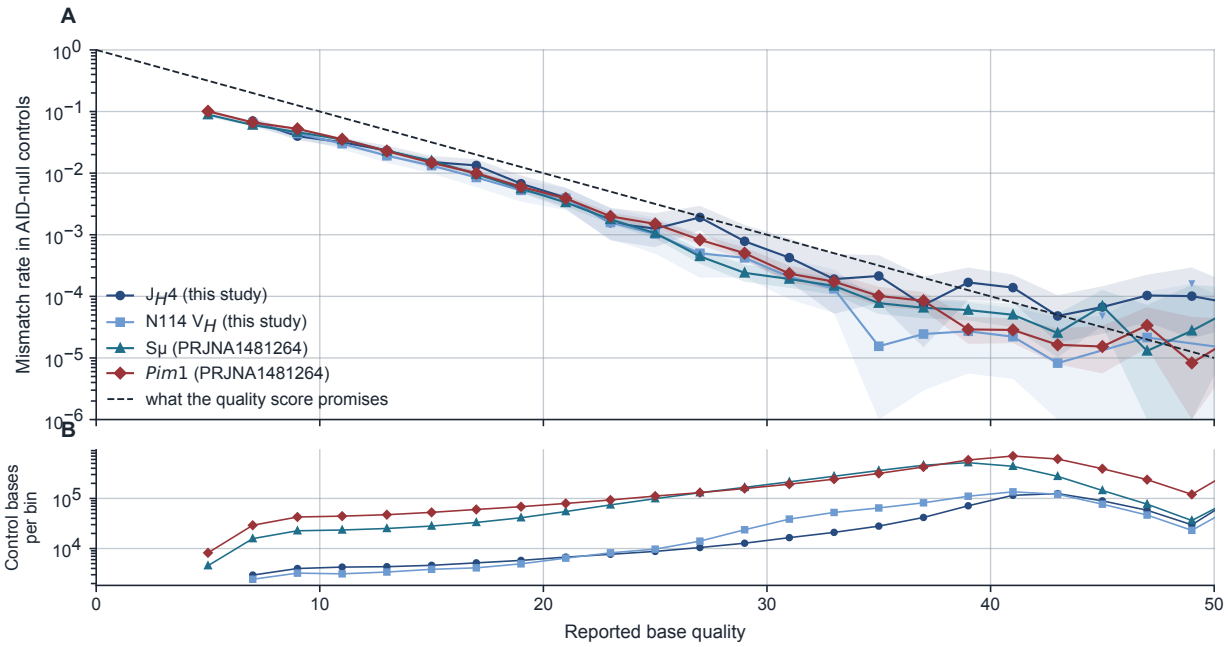
Burden, signature and indels are clearer; E is split into calibrated-call and raw-mismatch tracks.

Figure S1. Experimental design



Experimental design separated from computational processing.

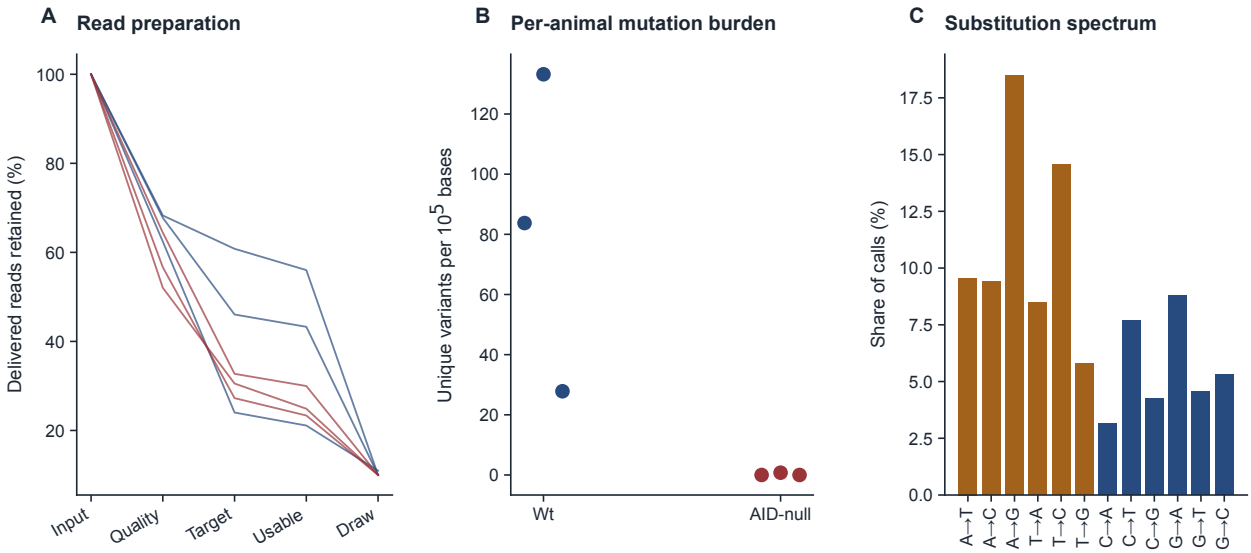
Figure S2. Reported quality and observed control error



Shading: 95% binomial intervals. Triangles: upper bounds with zero mismatches. Bins require ≥2,000 bases.

Aligned support panel; explicit interval and zero-mismatch definitions.

Figure S3. Representative project outputs

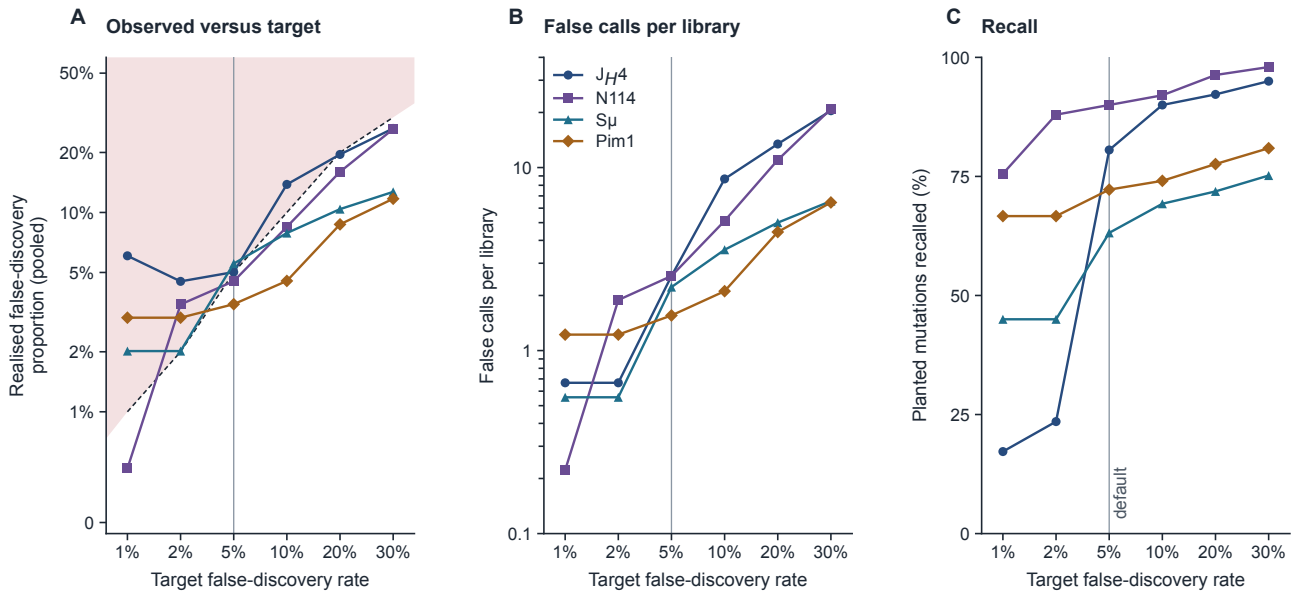


Example outputs from the same J_H4 project: one line per library in A, one point per mouse in B.

The report also includes positional profiles, hotspot use, clone summaries, and downloadable tables.

Three readable outputs replace six small instruction cards.

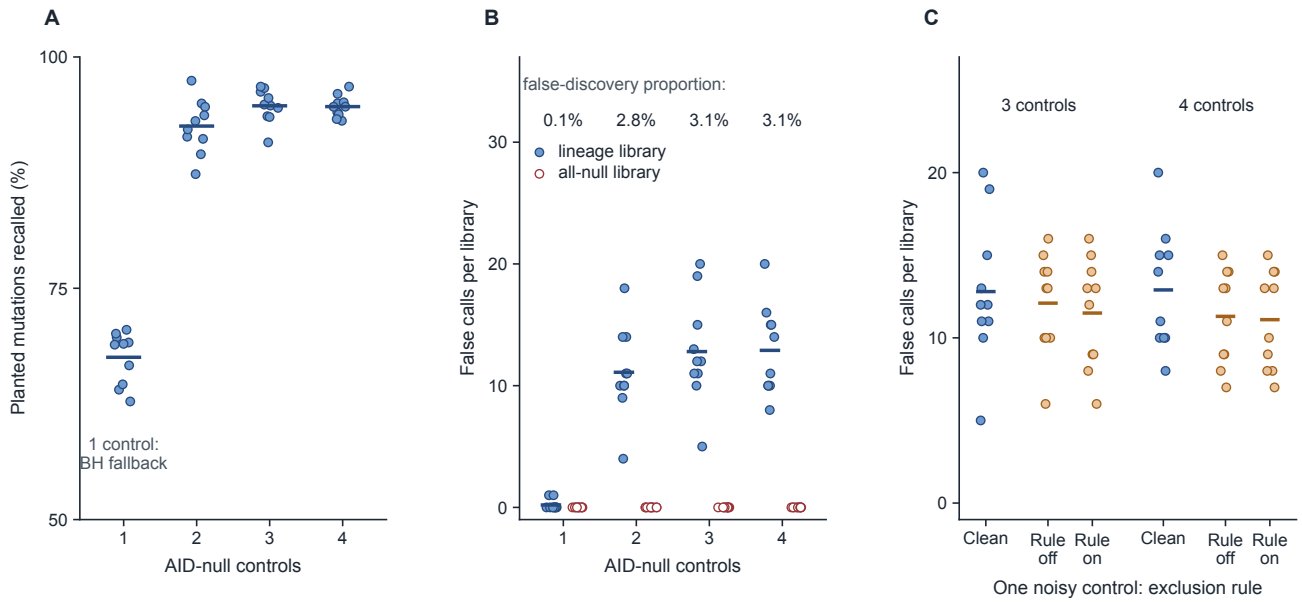
Figure S4. Empirical calibration against planted truth



A: pooled false-discovery proportion; linear to 1%, logarithmic above. Vertical line: 5% default target.

True zero FDP retained; linear-to-log scale explained; default target marked.

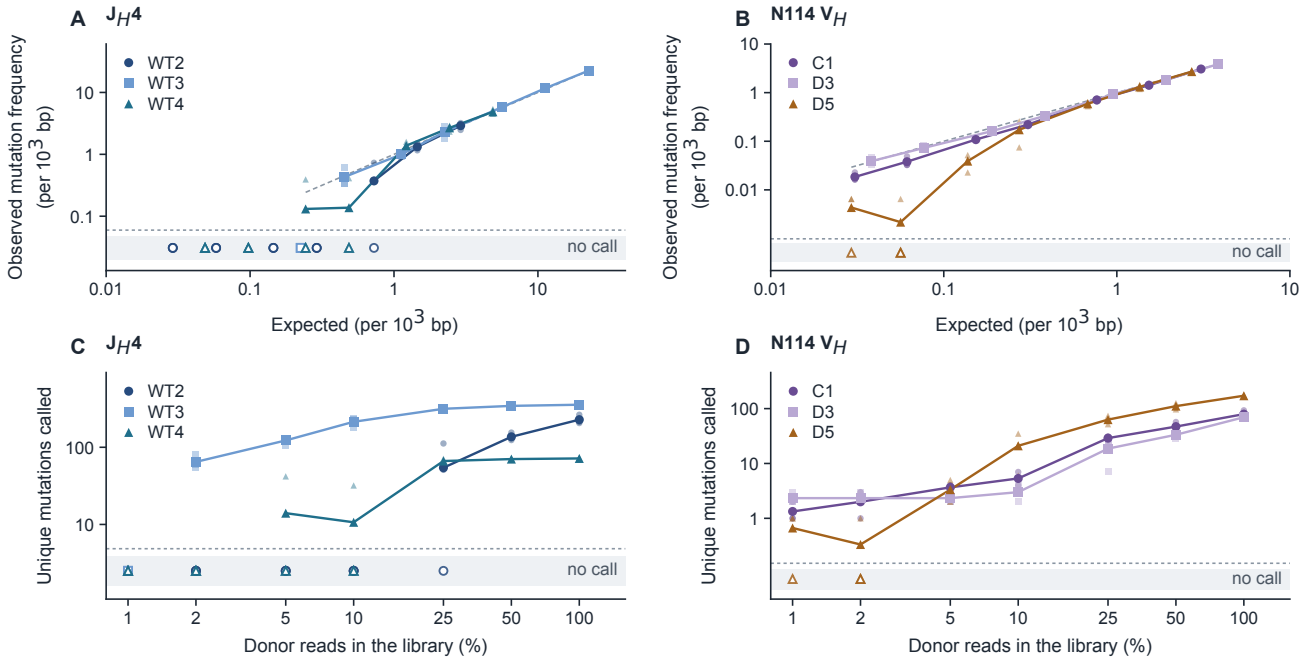
Figure S5. Control number and noisy-control handling



One-control designs use a BH fallback; two or more controls support empirical decoy calibration.

BH fallback distinguished from decoy calibration; conditions named explicitly.

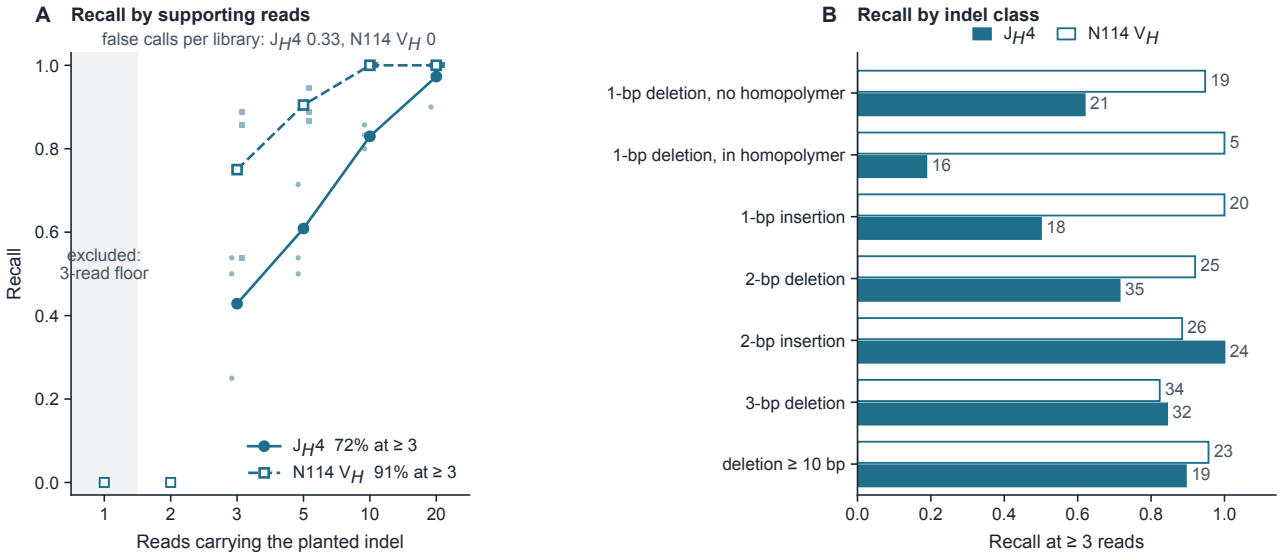
Figure S6. Dilution into AID-null reads



Dashed diagonal: expected = observed. Lower strips mark zero calls; curves retain the original library identities.

More room for axes; zero-call and identity-line explanations retained.

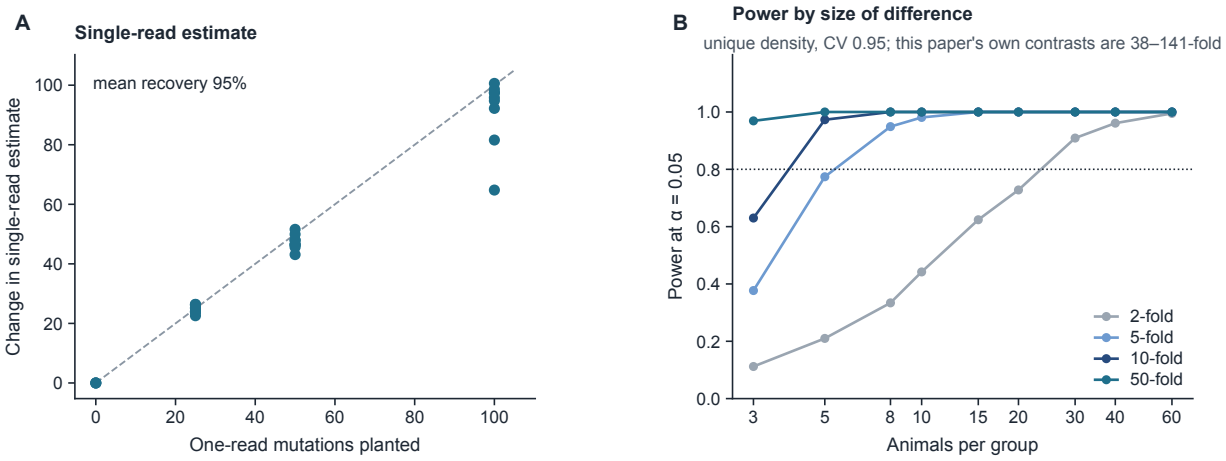
Figure S7. Indel spike-ins



Shaded support levels are excluded by the ≥3-read rule. Numbers beside class bars count planted indels.

Exclusion rule and planted-event counts stated explicitly.

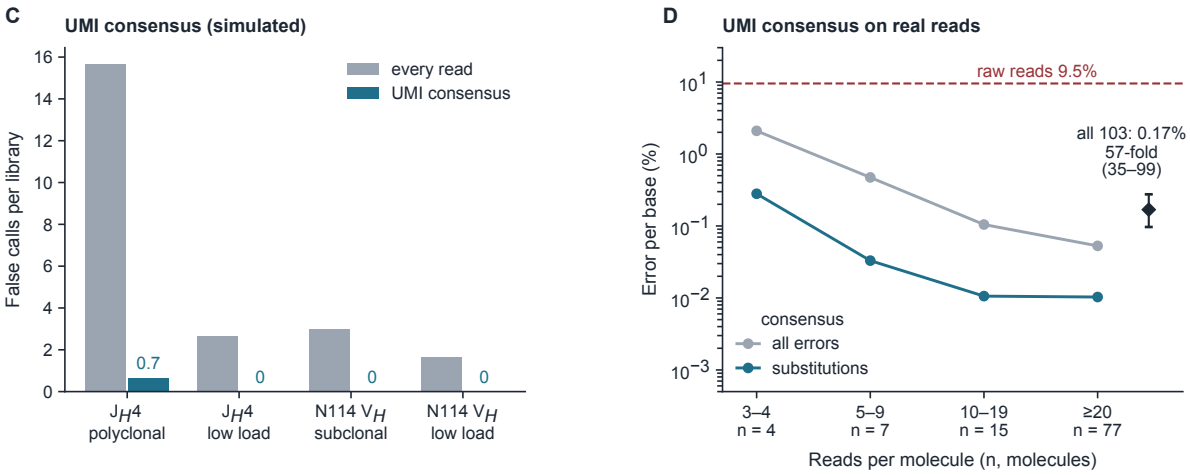
Figure S8. Estimation and pilot-based power (A–B)



A–B: estimation and pilot-based power.

Estimation/power separated from UMI panels under the same figure number.

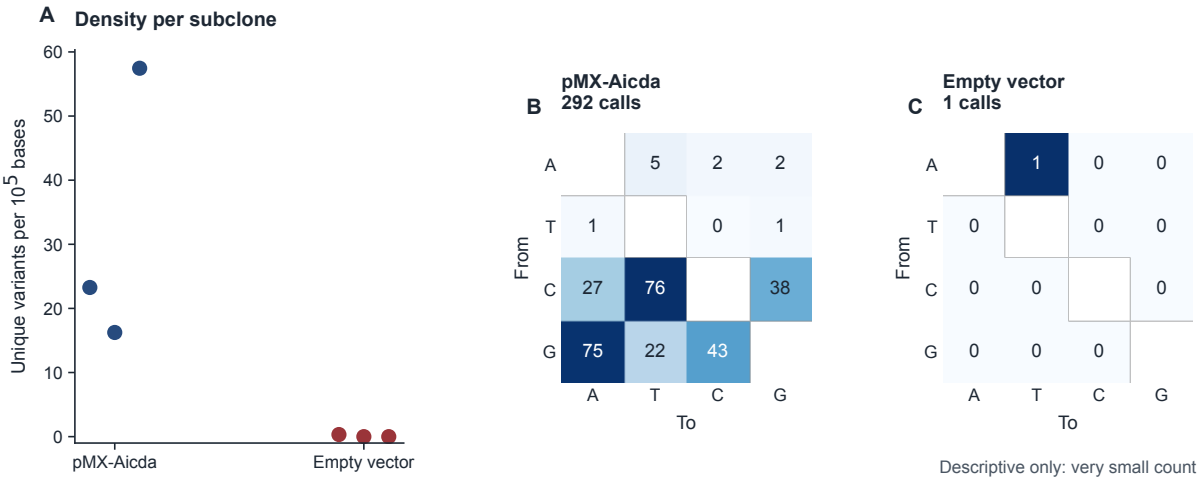
Figure S8. UMI consensus (C–D)



C–D: simulated and measured UMI consensus performance; n denotes molecules.

A separate page preserves readable labels and molecule counts.

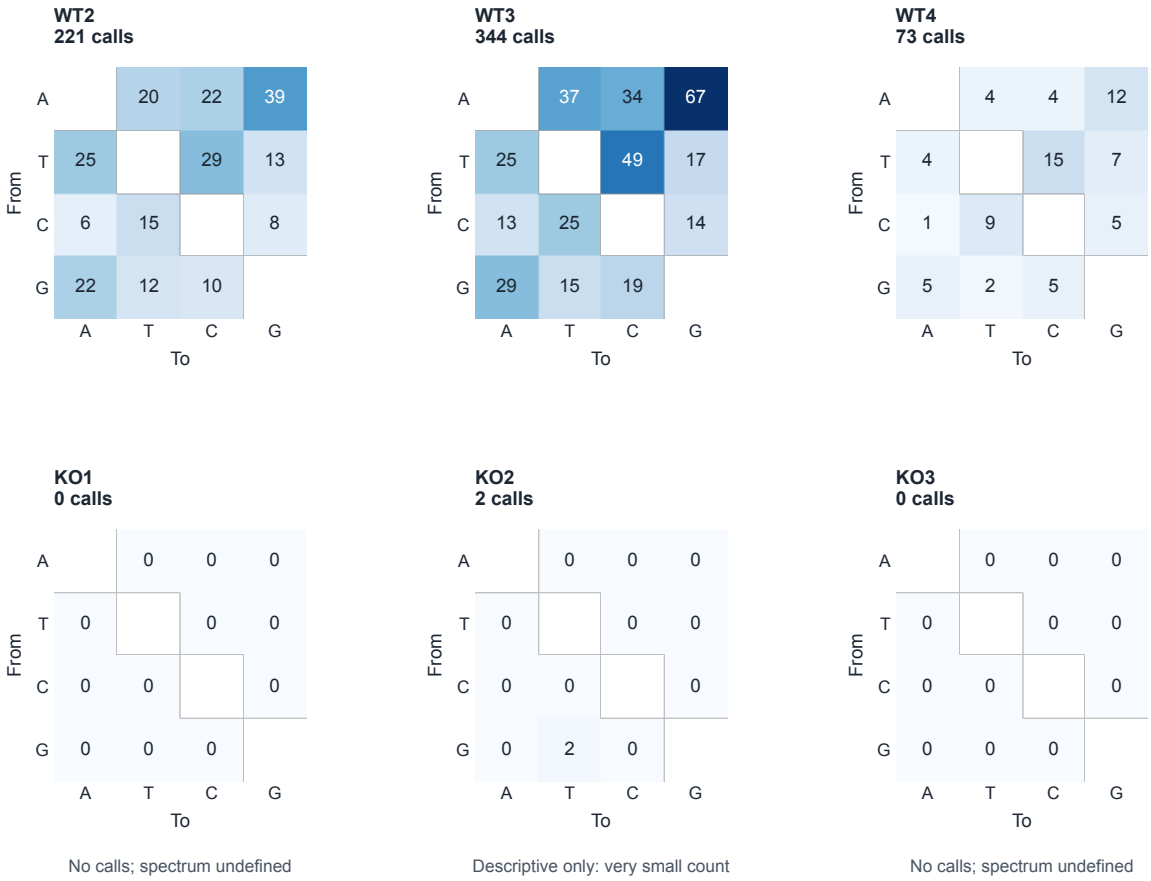
Figure S9. N114 results: pooled view



Matrices show absolute unique-variant counts. Empty-vector calls are too few for a stable percentage spectrum.

Absolute counts replace unstable control percentages; density includes all six subclones.

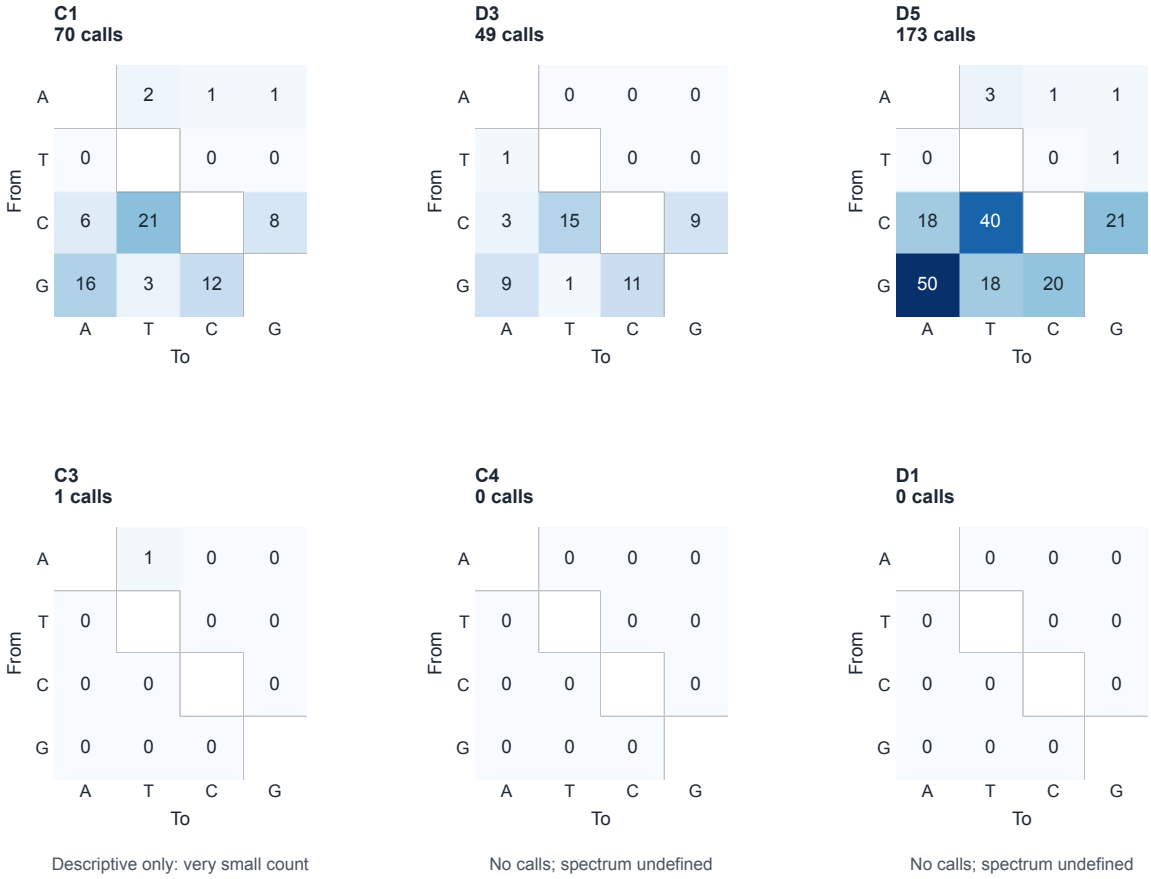
Figure S9. Per-library JH4 spectra



J_H4: absolute counts, one matrix per library. All panels use the same colour scale; diagonal cells are not substitutions.

Common absolute-count colour scale; zero-event spectra marked undefined.

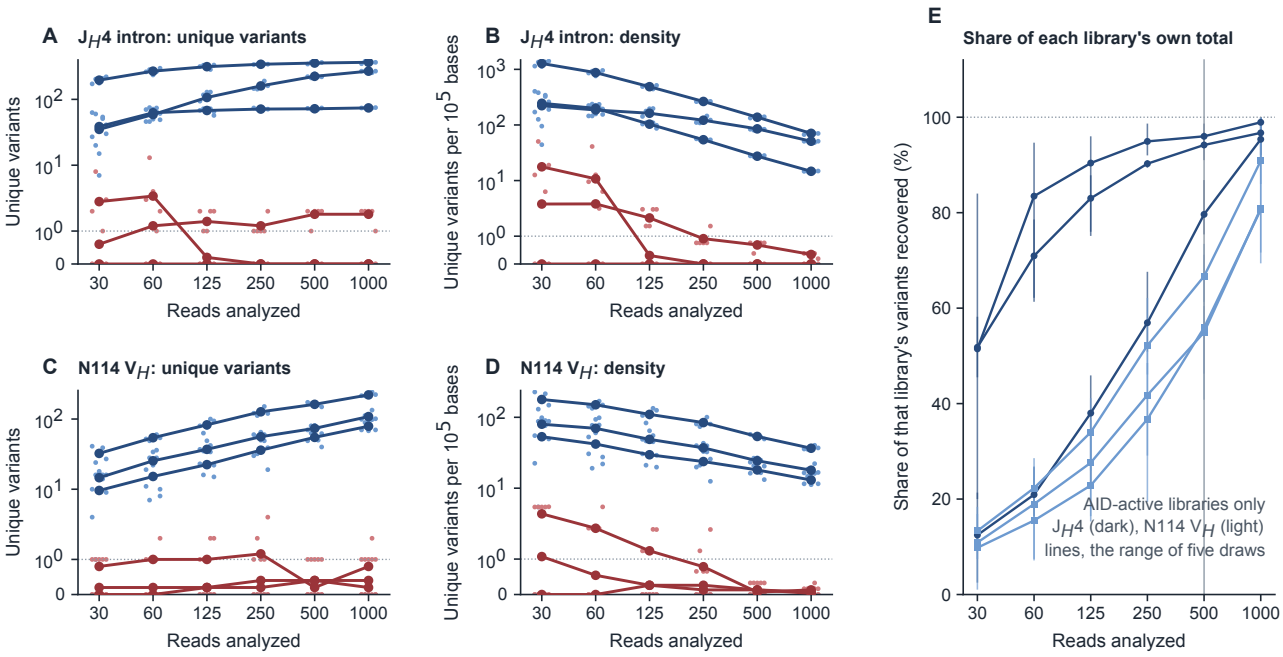
Figure S9. Per-library N114 spectra



N114 V_H: absolute counts, one matrix per library. All panels use the same colour scale; diagonal cells are not substitutions.

Common absolute-count colour scale; tiny totals explicitly qualified.

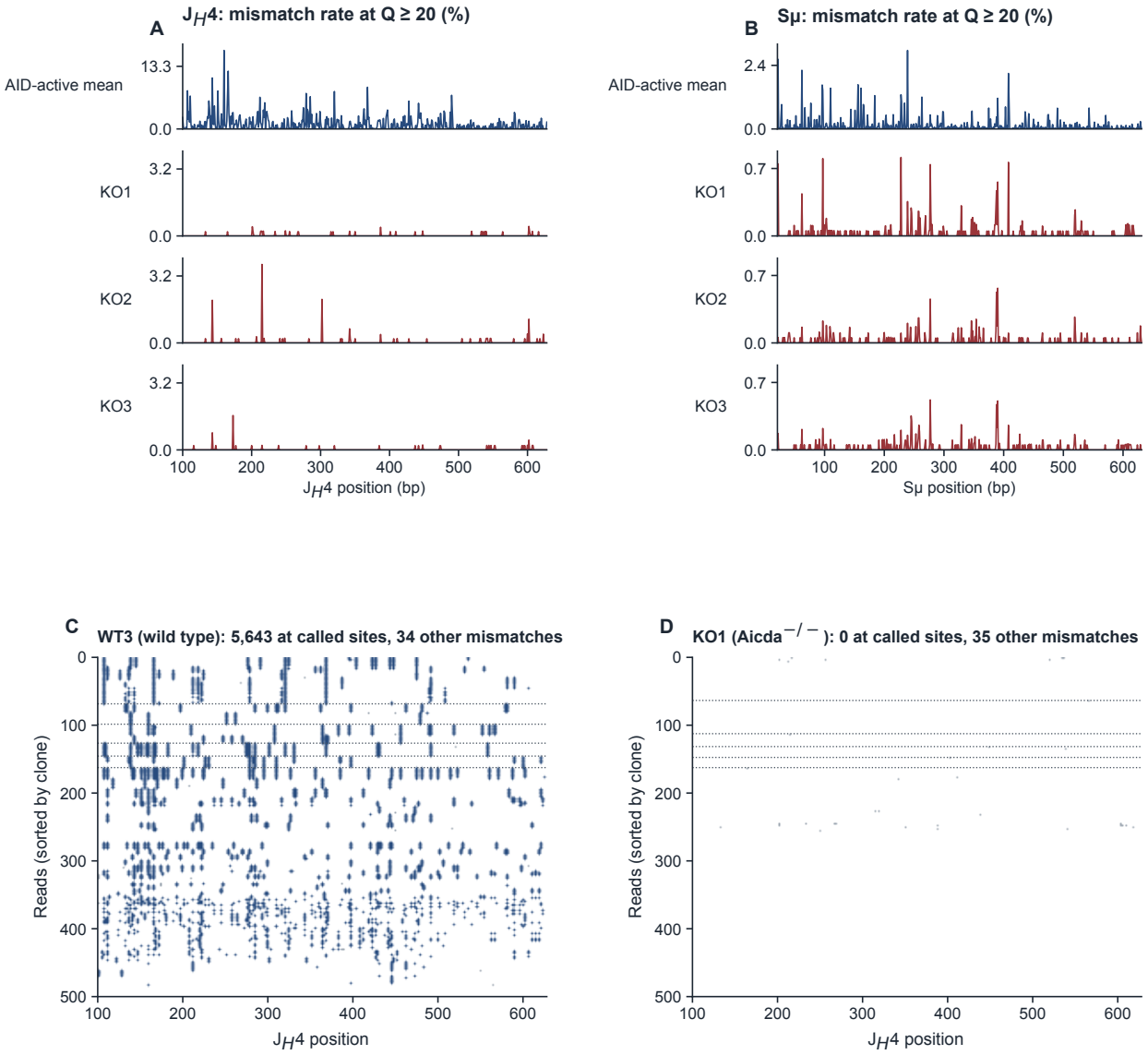
Figure S10. Sequencing depth and discovery



Recovery in E uses each library's maximum discovery across all depths/draws; lines retain library identity.

Clarifies the library-specific maximum-discovery denominator.

Figure S11. Positional background and individual reads

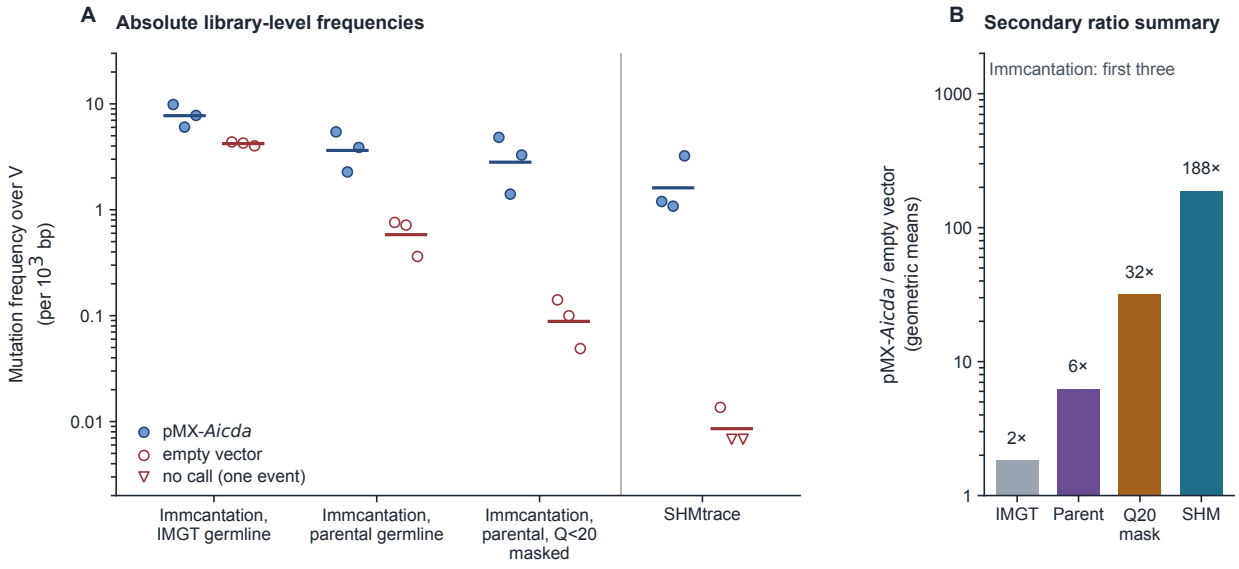


A–B: controls have separate tracks with a shared scale within each locus; the AID-active mean uses its own scale.

C–D: individual J_H4 reads, ordered by clone and mutation load. Colours mark called sites; grey marks other mismatches.

Separate control tracks replace overlaid profiles and small insets.

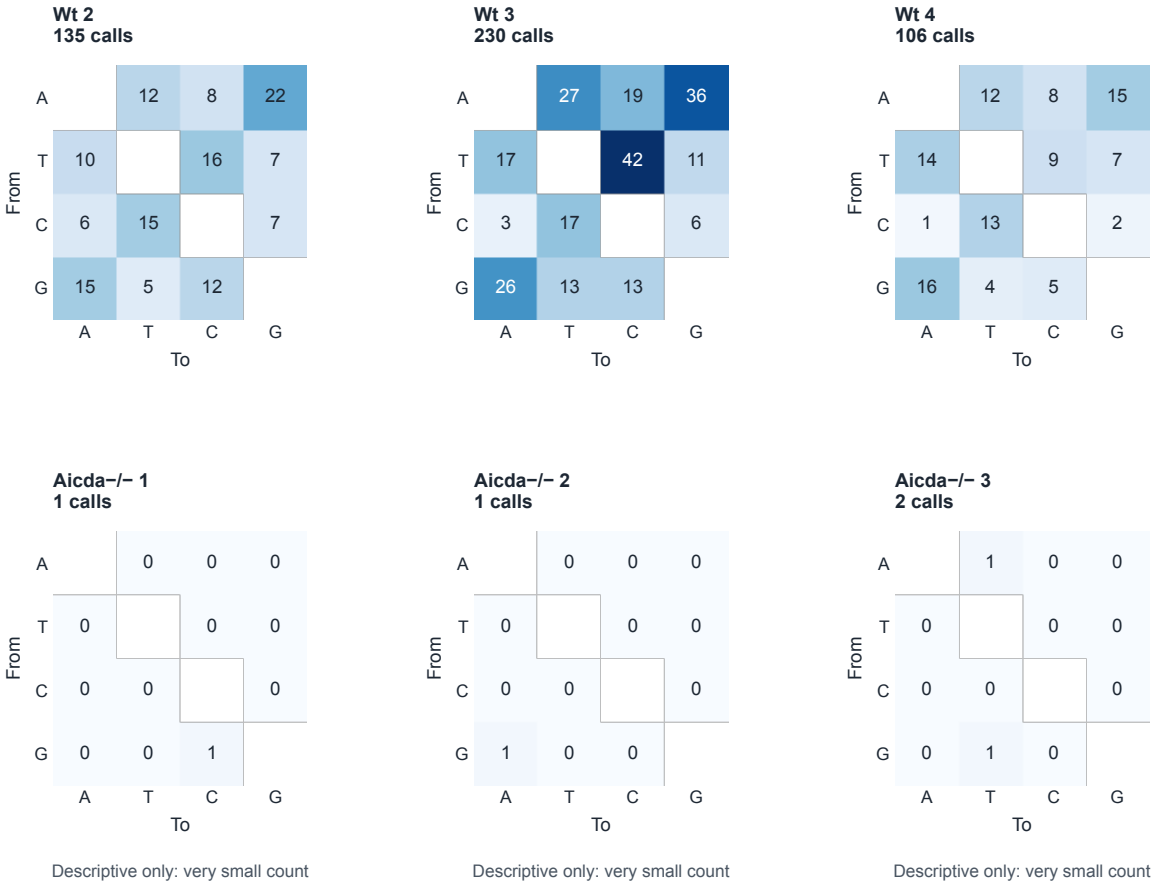
Figure S12. Immcantation comparison



Routes differ in germline reference and quality masking. Ratios use a one-event floor for zero-call libraries.

Absolute values lead; ratios remain a secondary summary with an explicit zero floor.

Figure S13. Sanger calls by mouse



Descriptive only: very small count

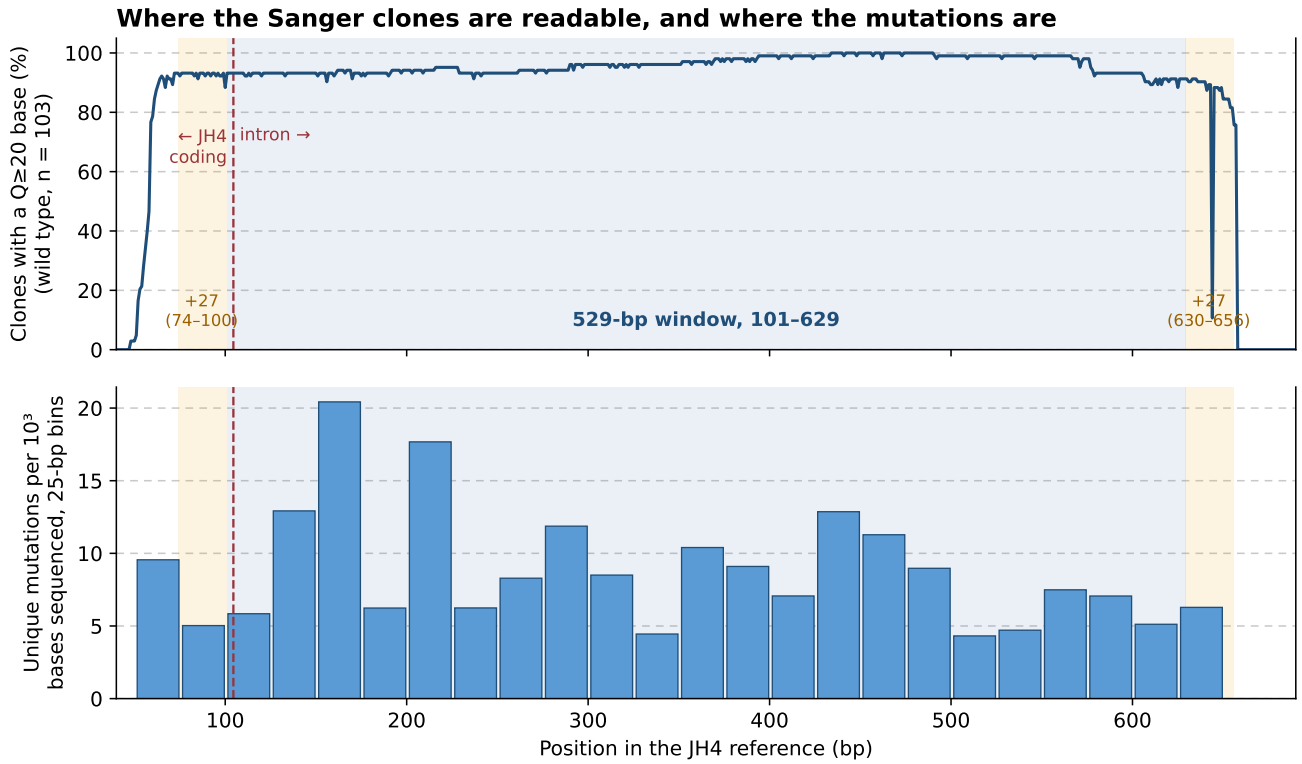
Descriptive only: very small count

Descriptive only: very small count

Sanger calls at Q ≥ 20. Absolute counts share one colour scale; tiny control totals are not interpreted as spectra.

Counts share one scale; small control totals are descriptive only.

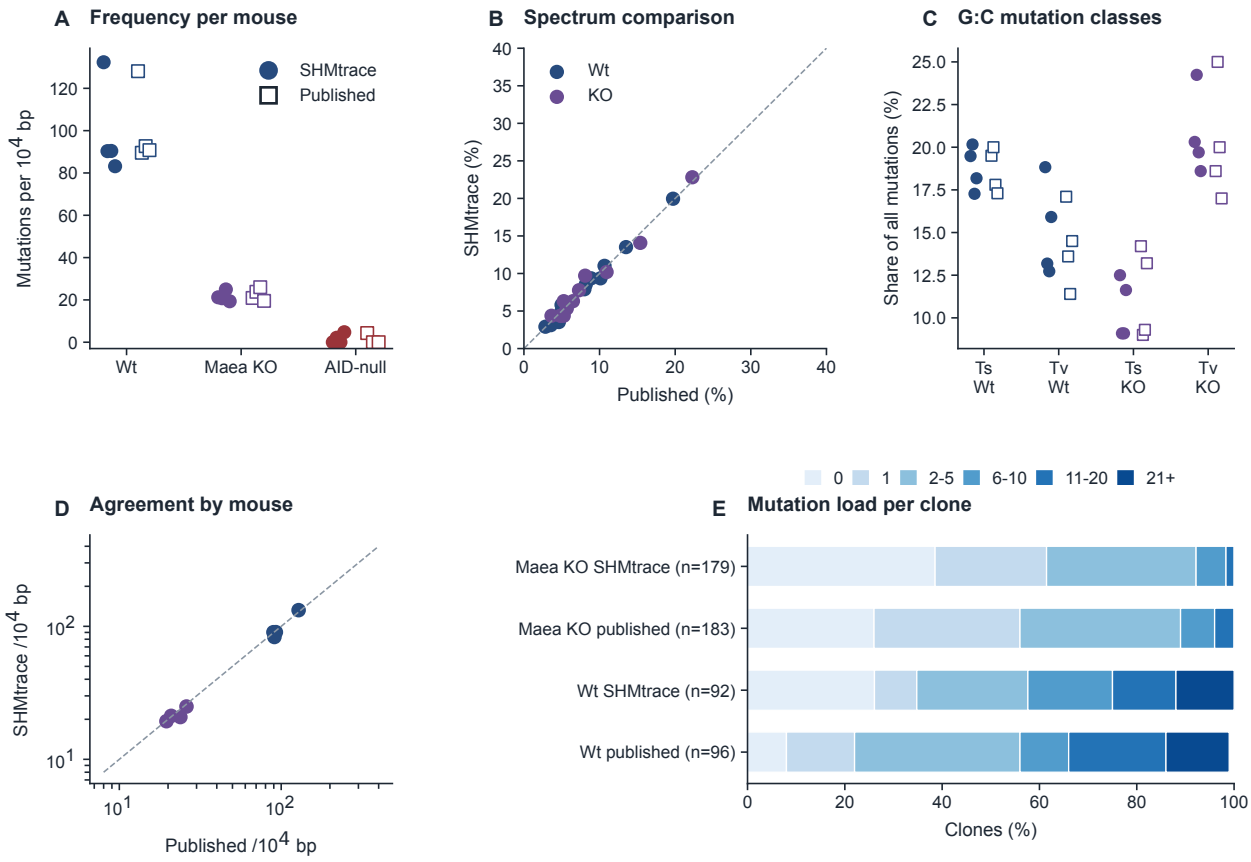
Figure S13. Readable Sanger window



Coverage: Q >= 20 bases per clone. Density: unique events / observed bases within each 25-bp bin.

Saved vector curves retained; denominator clarification added without reprocessing traces.

Figure S14. Maea Sanger reanalysis



Filled circles: reanalysed; open squares: published. Detailed test reproduction and provenance are in the companion table.

Larger data panels; numerical reproduction table moved to a companion page.

Figure S14. Published and reanalysed results

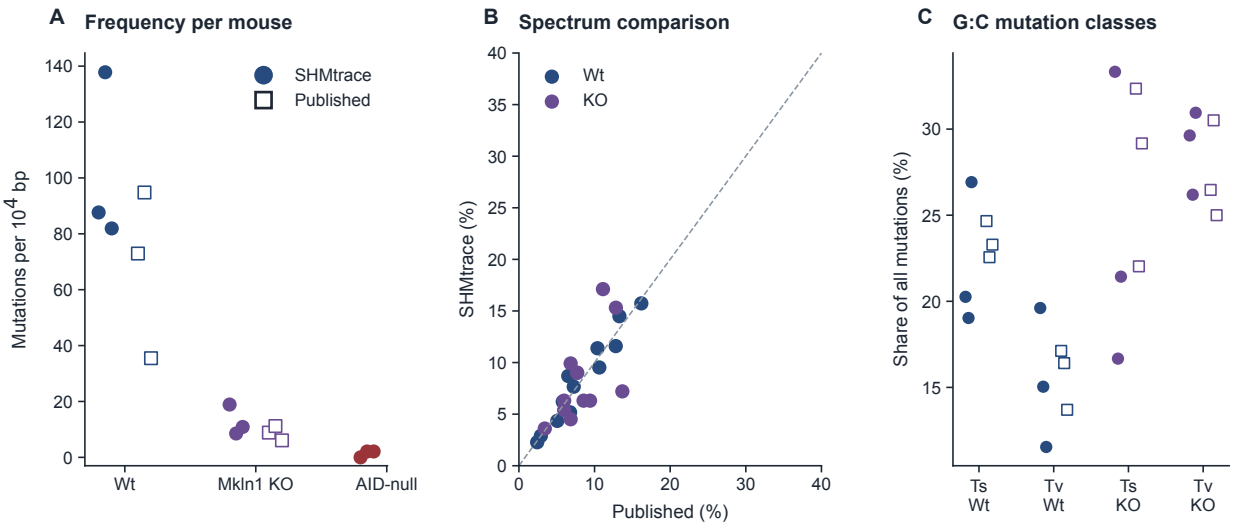
Figure S14: published and reanalysed results

Measure	Published	SHMtrace
Wt / Maea frequency	4.4×	4.6×
P (one-tailed t)	0.014	0.0002
G:C Ts P	0.0028	0.0003
G:C Tv P	0.0285	0.025
Mice, Wt / Maea / null	4 / 4 / 3	4 / 4 / 4
Clones, Wt and Maea	284	271
Background explains, Wt	–	1-2%
... Maea	–	7-8%
Lineages under 2+ mice	–	14 (49 clones)
Analysis	SHMTool	16 s

Published and reanalysed tests can differ in included sequences or definitions; see STAR Methods and the source-study notes.

Same numerical table, separately readable and exported as CSV.

Figure S15. Mkln1 Sanger reanalysis



Filled circles: reanalysed; open squares: published. Detailed test reproduction and provenance are in the companion table.

Larger burden, spectrum and class plots; table separated.

Figure S15. Published and reanalysed results

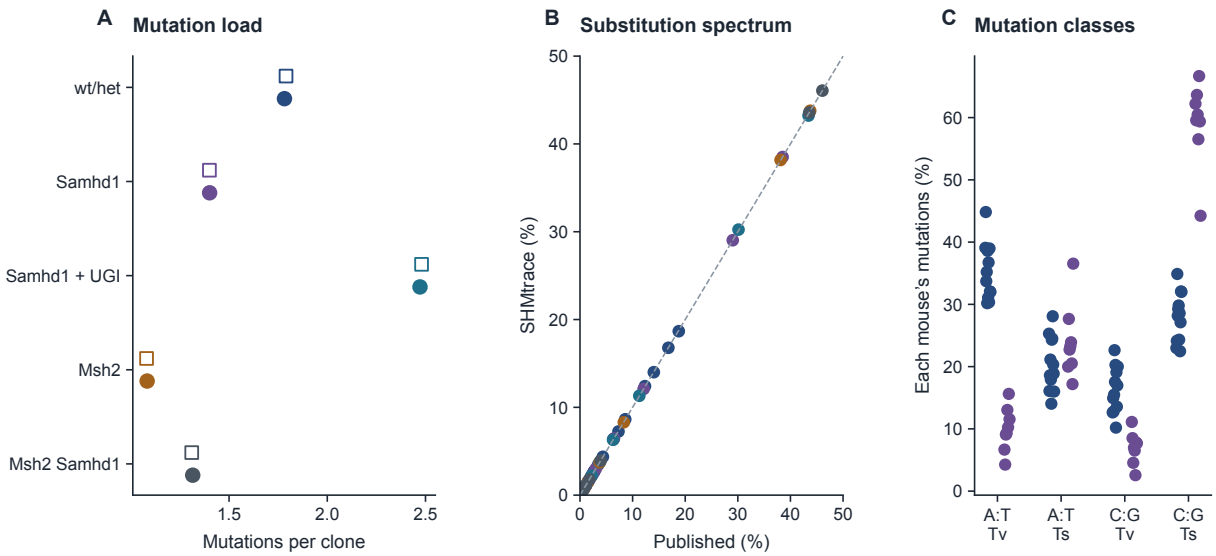
Figure S15: published and reanalysed results

Measure	Published	SHMtrace
Wt / Mkln1KO	7.8×	8.0×
P (one-tailed t)[b]	0.0137	0.0038
G:C Ts P	0.23	0.77
G:C Tv P	0.004	0.008
... S2C bars[a]	0.013 / 0.31	–
Clones, Wt / KO	110 / 227	88 / 193
Paralog clones out	–	85
Background explains, Wt	–	2-3%
... Mkln1KO	–	12-30%
Analysis	Benchling	15 s

a: Ts/Tv as drawn in the paper's Fig. S2C. b: published column computed from its tables. The KO spectrum comparison uses the published KO table traceable to folder K2.

Same numerical table with source-study footnotes and CSV.

Figure S16. SWHEL sequence reanalysis



2,804 of 2,809 published mutations recovered across 1,795 clones and 40 animals; percentages are descriptive summaries.

Filled circles: SHMtrace; open squares: published. Detailed effects and differing tests are in the companion table.

Near-unity correlation precision removed; recovery counts made explicit.

Figure S16. Published and reanalysed effects

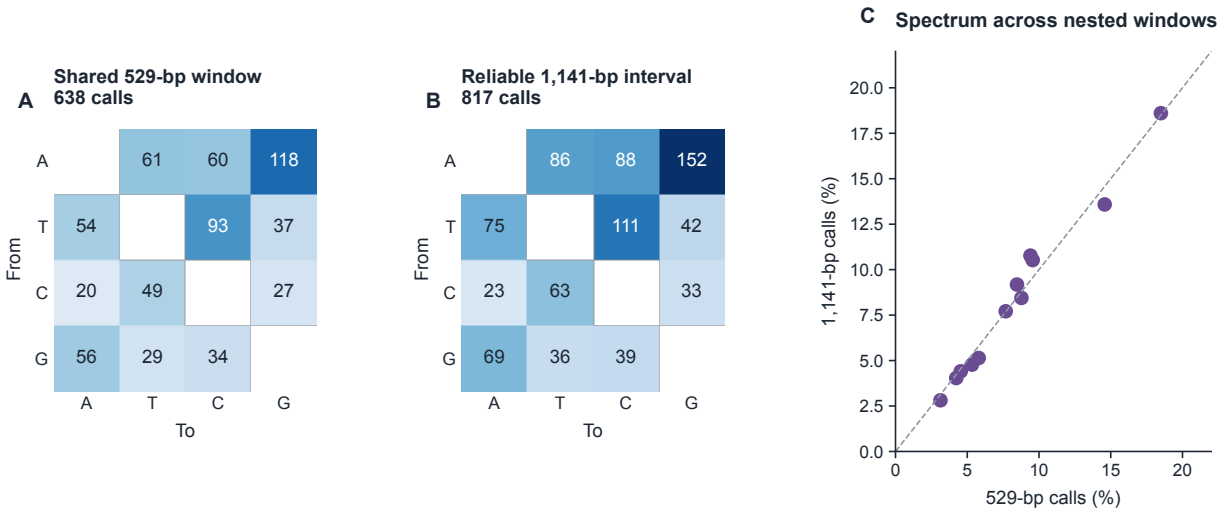
Figure S16: published and reanalysed effects

Measure	Published	SHMtrace
A:T transversions	down ****	down ****
A:T transitions	unchanged	unchanged
C:G transversions	down ****	down ****
C:G transitions	up **	up ****
Total mutation[a]	lower, ns	lower ****

The published total-mutation comparison used one-way ANOVA; SHMtrace uses Welch on log frequency. Other tests retain their original definitions.

Effect comparison and differing test definitions moved to a readable table.

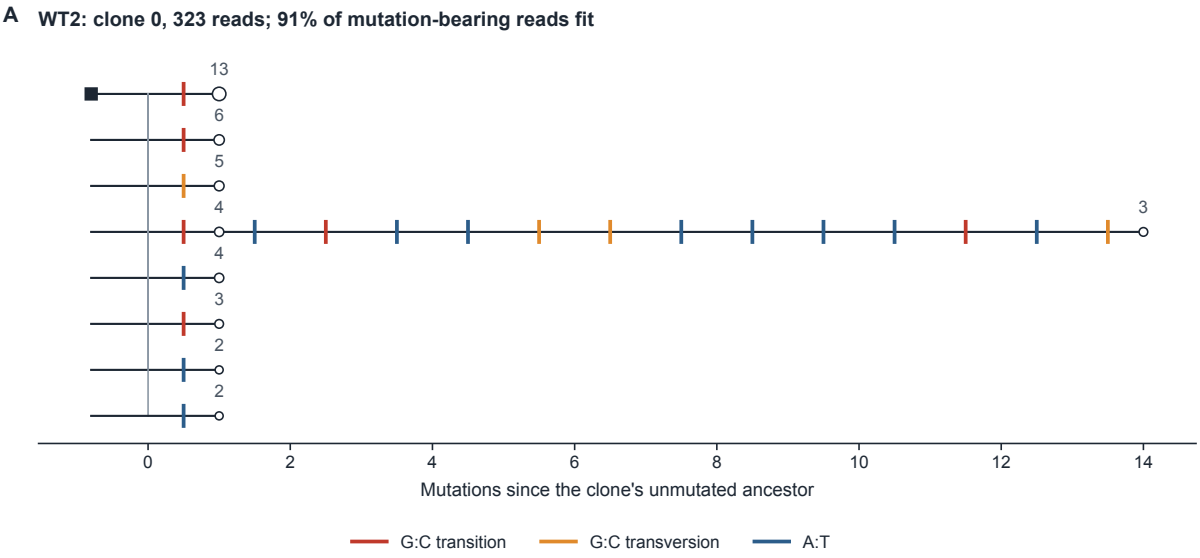
Figure S17. Spectra across nested windows



The shared window is contained within the full interval: this is a descriptive comparison of nested data, not independent validation.

Absolute counts and an explicit qualification that windows overlap.

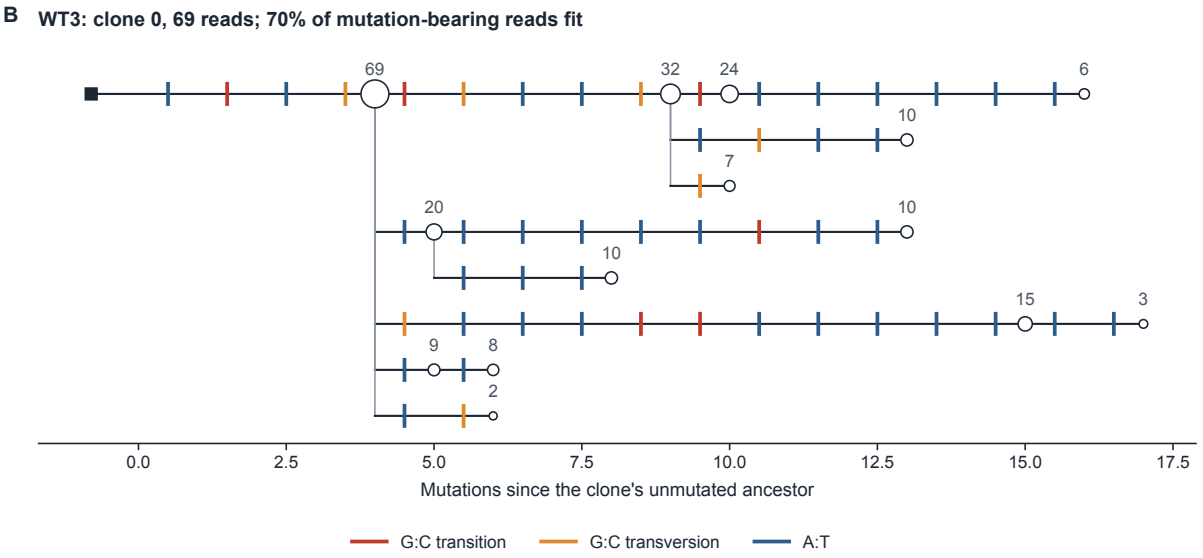
Figure S17. WT2 clone 0 reconstruction



Node labels are nested read counts; ticks are mutations. Branches represent reconstructed sequence relationships, not observed ancestry.

Full-width tree; fit is scoped to mutation-bearing reads.

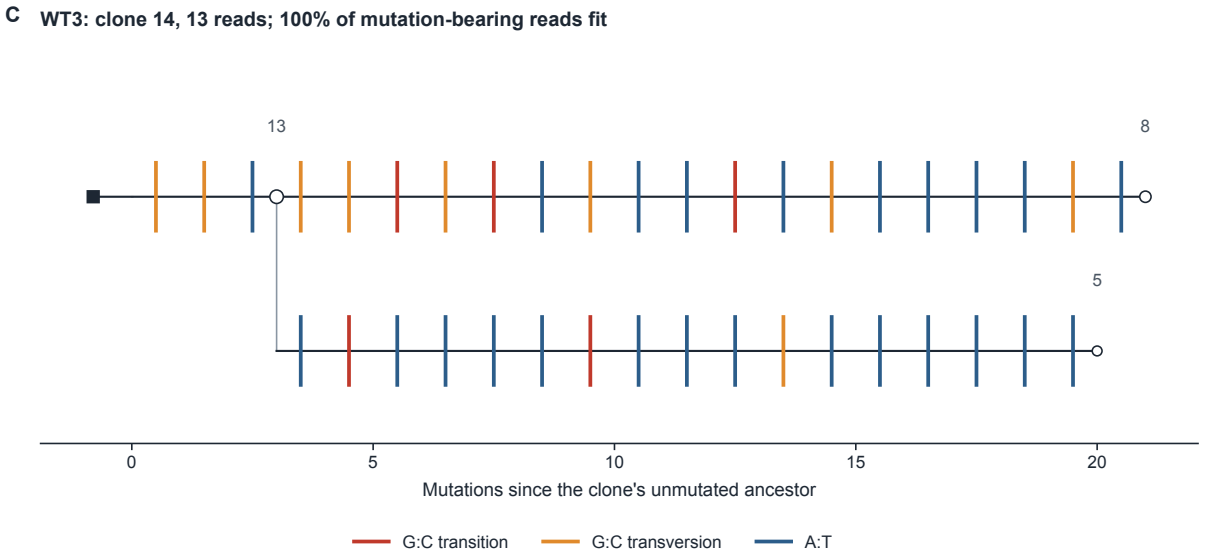
Figure S17. WT3 clone 0 reconstruction



Node labels are nested read counts; ticks are mutations. Branches represent reconstructed sequence relationships, not observed ancestry.

Full-width tree; read counts and reconstructed relationships explained.

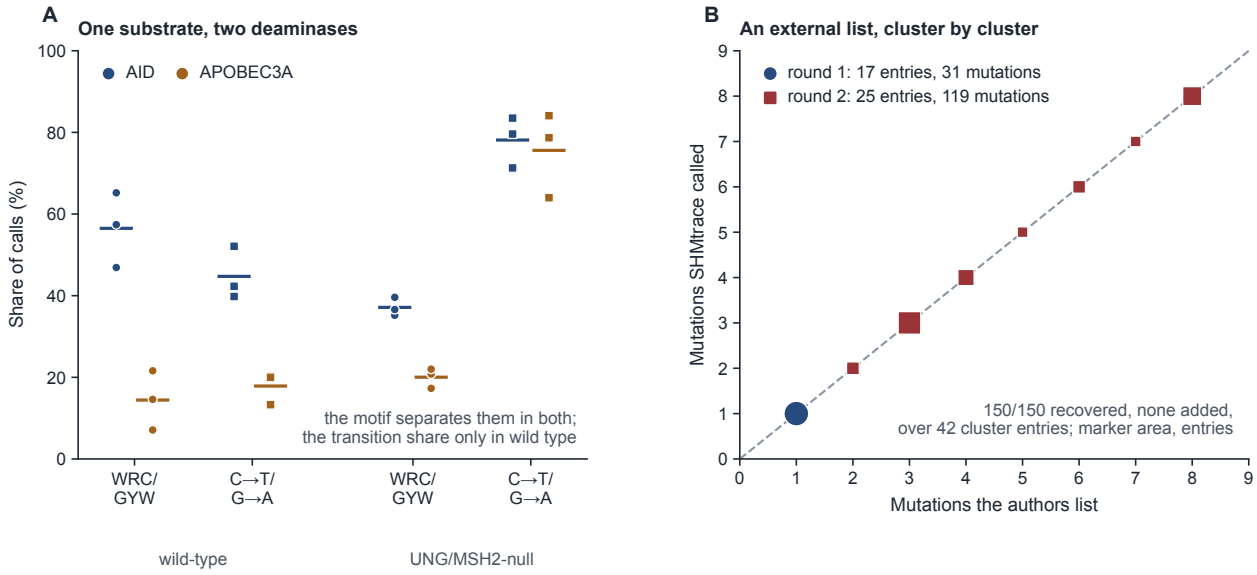
Figure S17. WT3 clone 14 reconstruction



Node labels are nested read counts; ticks are mutations. Branches represent reconstructed sequence relationships, not observed ancestry.

Preserves the actual third example rather than incorrectly describing it as WT4.

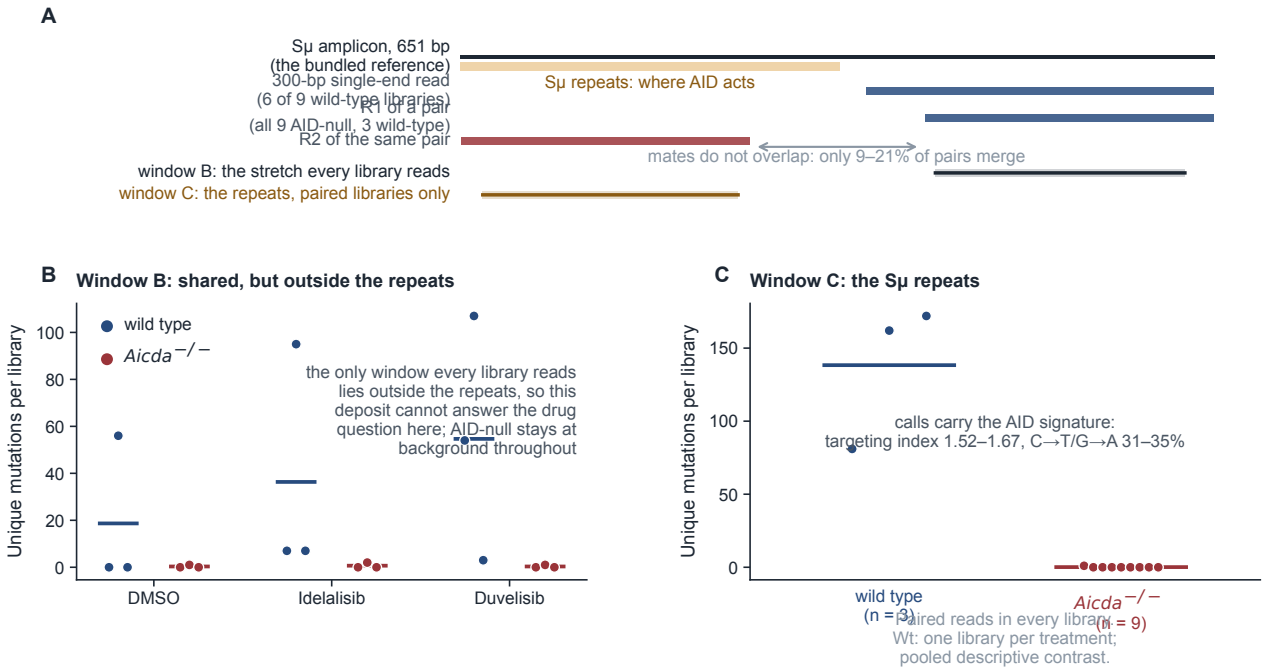
Figure S18. Engineered diversification



A: distinct deaminases on one substrate. B: recovery of the external cluster list; marker area encodes entries.

Local validation questions and external-list recovery remain explicit.

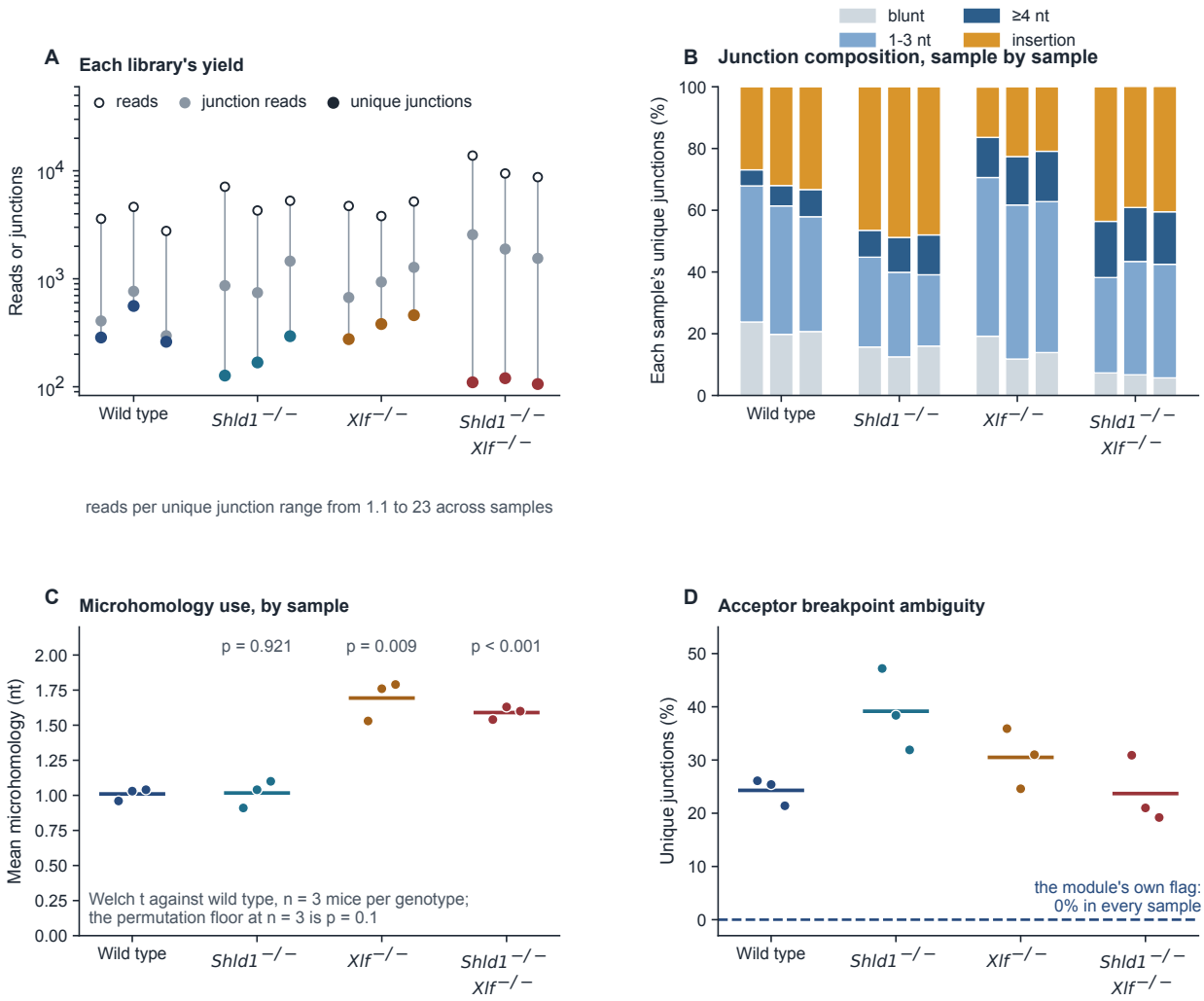
Figure S19. Public Sμ libraries



B uses the interval shared by all libraries; C uses the repeat interval in paired-read libraries only.

More separation between shared-window and repeat-window comparisons.

Figure S20. PacBio switch junctions



reads per unique junction range from 1.1 to 23 across samples

Each dot/bar is a sample. D measures placement ambiguity directly, separately from microhomology length.

One composition bar per biological sample replaces group-mean stacks.

Figure S21. Platform measurements

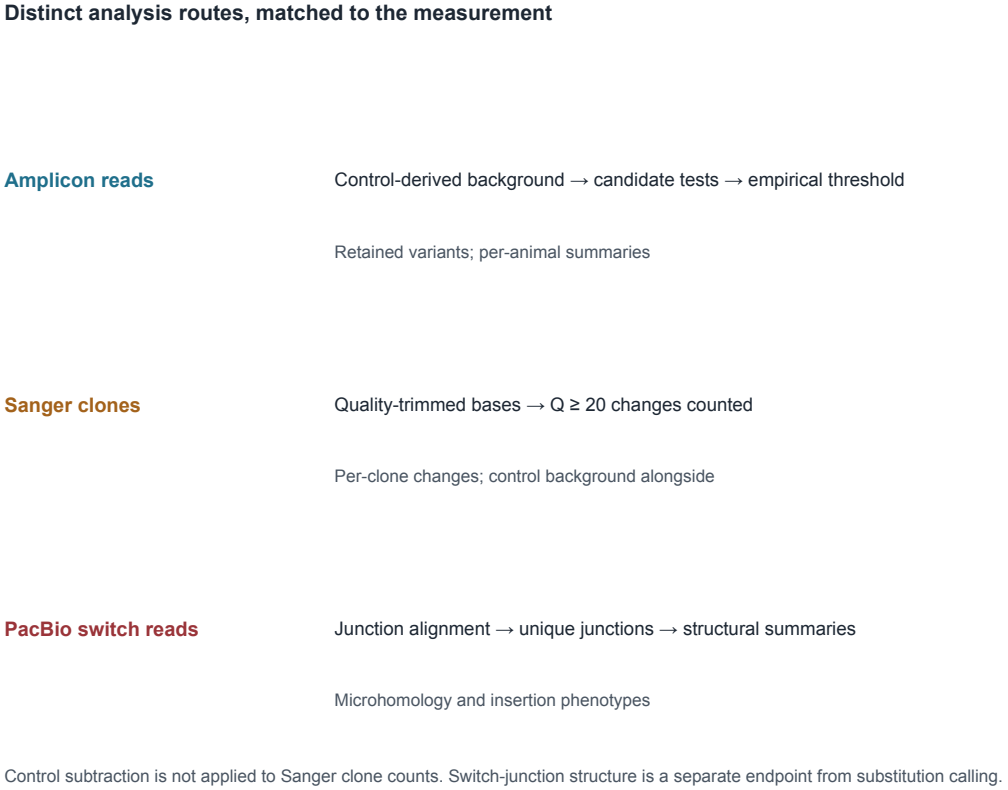
What each platform measures

Platform	Measurement unit	Analysis route	Output / qualification
Sanger	Cloned amplicon	Q ≥ 20 changes counted; background reported alongside	Per-clone load and spectrum; shared readable window
Nanopore	Amplicon read	Control-calibrated variants; optional UMI consensus	Long positional span; junction-defined clones
Illumina	Merged paired read	Control-calibrated variants; reference and layout stated	Mutation burden and signature; span limited by read layout
PacBio	Switch-junction read	Junction alignment and unique-junction summaries	Microhomology, insertions; placement ambiguity checked

Measurement units and counting rules differ across routes; absolute densities should not be treated as interchangeable.

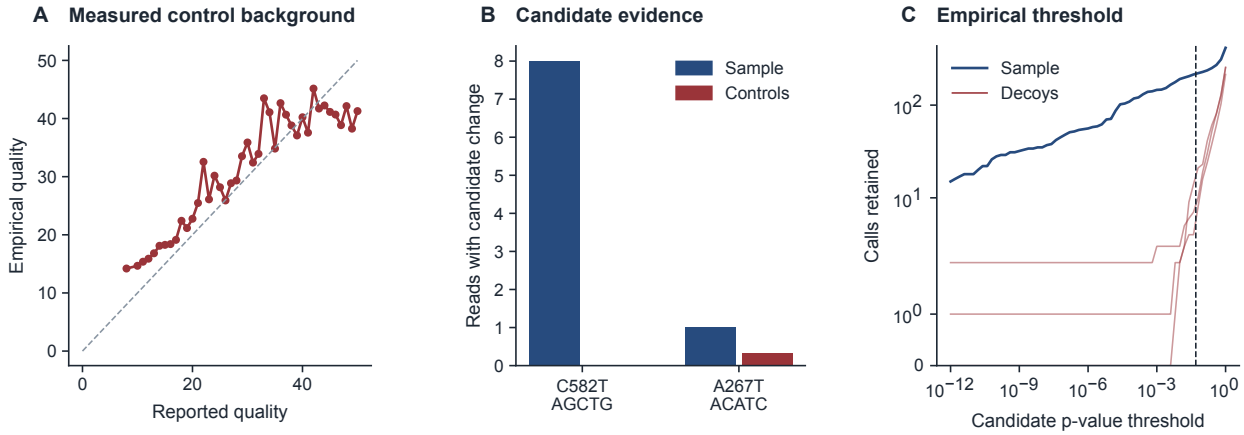
Compact comparison table replaces dense diagrams and prose cards.

Figure S22. Distinct measurement routes



A route-comparison plate replaces the redundant workflow overview.

Figure S23. Background, testing and empirical calibration



Estimate background

Pool by substitution, strand, and quality; shrink position-level estimates toward pooled rates.

Test candidates

Compare observed support with control-derived expected support.

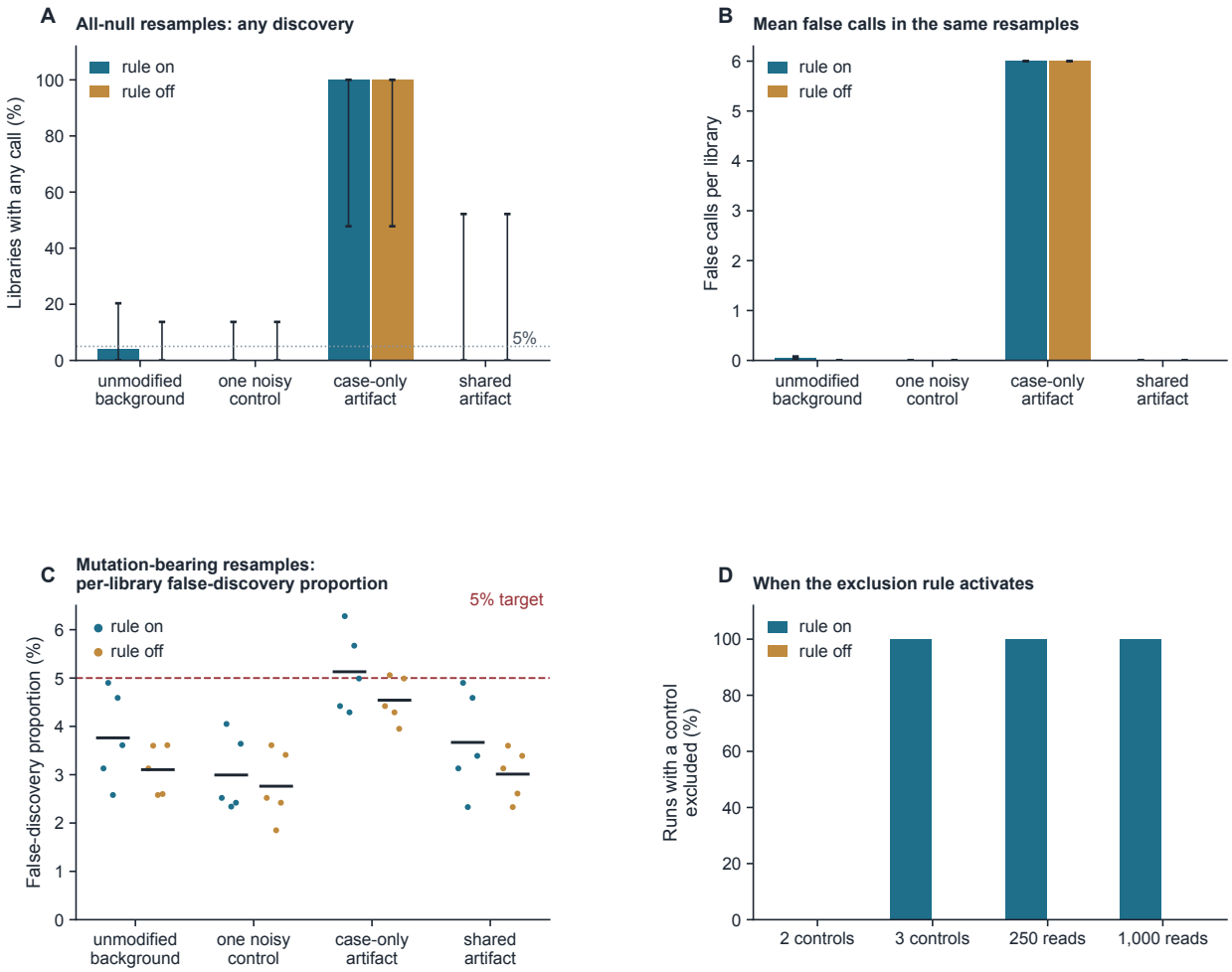
Calibrate selection

Held-out control calls supply decoys; select an empirical threshold.

J_H4 example from the saved statistical-engine figure data. Preprocessing is documented separately in Figures 1 and S26.

Three readable statistical steps replace six formula cards.

Figure S24. Artifact stress tests

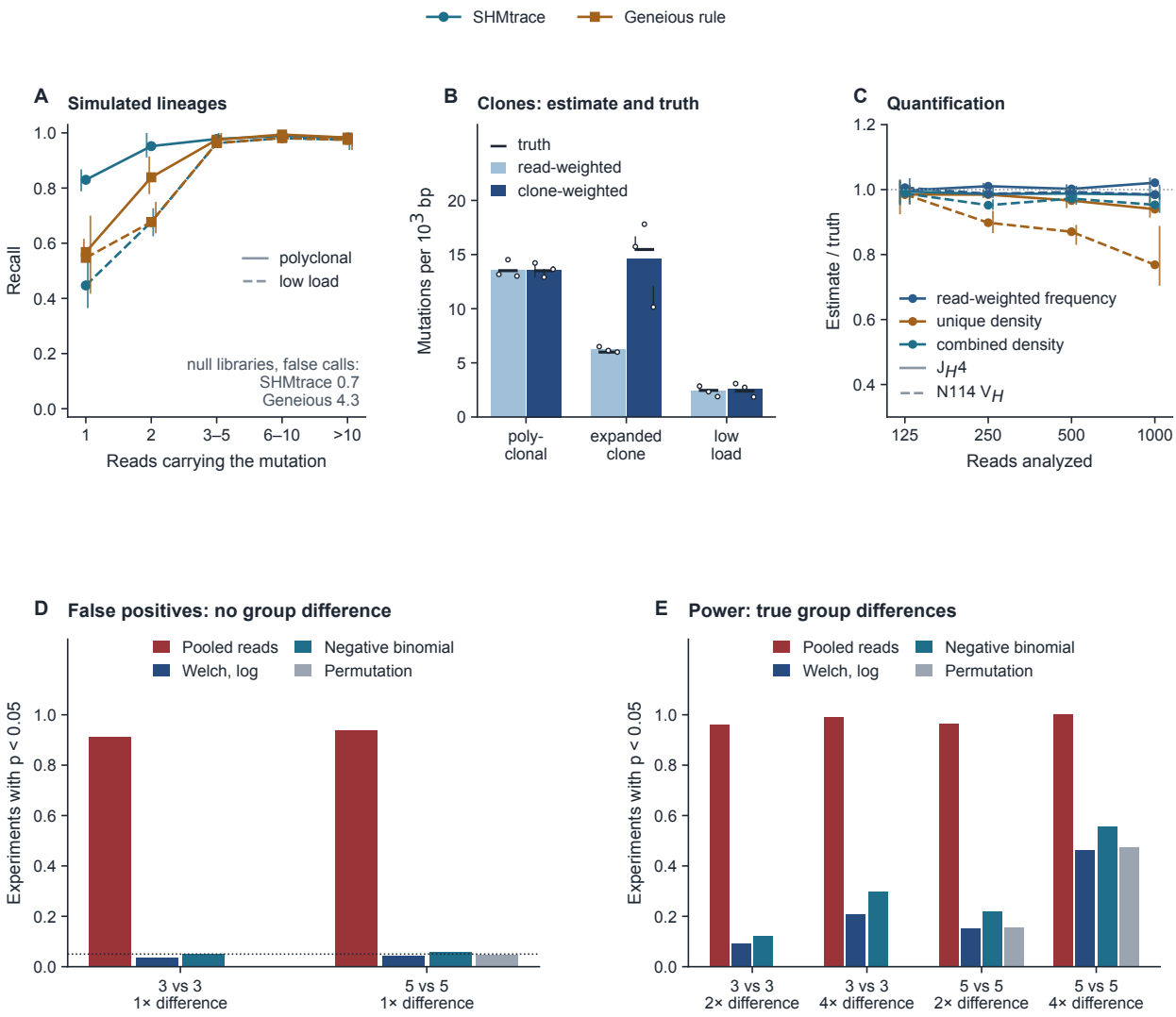


A: probability of ≥1 call in an all-null resample (95% exact binomial intervals). B: mean false calls (upper SEM).

C: per-library false / total calls with planted signal. Case-only artifacts illustrate a limit of matched-control calibration.

Any-discovery probability distinguished from per-library FDP; interval definitions explicit.

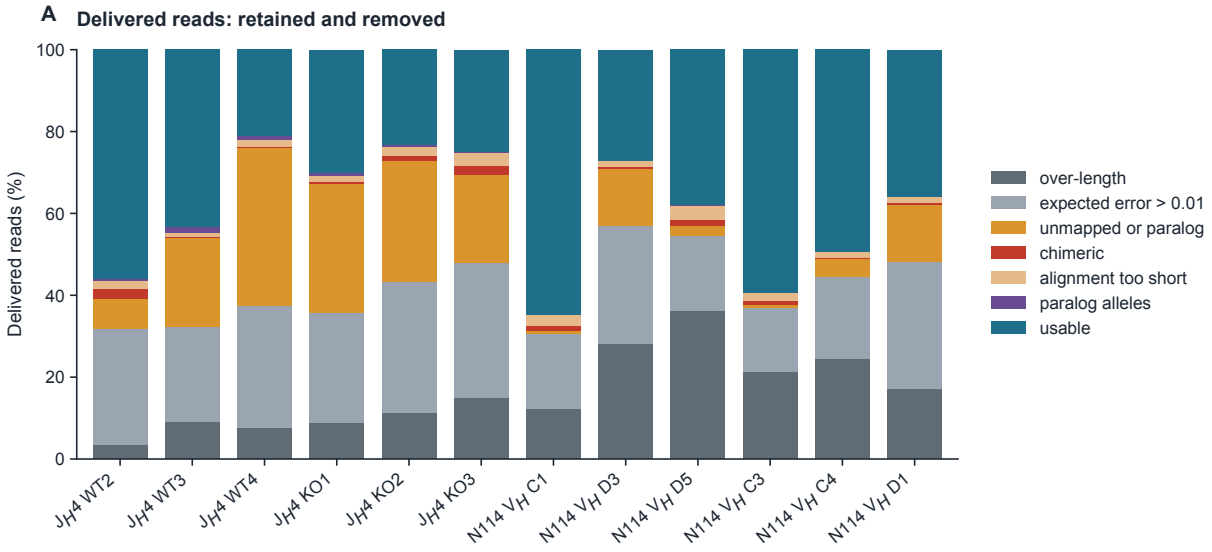
Figure S25. Downstream inference simulations



A–C: original simulation summaries and ranges. D–E: 300 experiments per condition; animals define groups.

Original Figure 3E–G retained as A–C; original H separated into false positives (D) and power (E).

Figure S26. Read accounting



One stacked bar per library; each segment is a recorded read-preparation outcome.

Original Figure 4I moved intact to a dedicated companion figure.